Collectively Big, Locally Small: Online Supplement

Systematic search, simulation study and coded corpus

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This supplement accompanies the survey *Collectively Big, Locally Small*. Section S1 documents the systematic database search summarized in Section 5.4 of the paper: sources, query strings, screening rules and their verification, the recall of the search against the coded corpus, the validation of the dictionary coder, and the field-level map. Section S2 gives the full design and results of the fifteen Monte Carlo simulations of the Graph Hierarchical Borrowing (GHB) blueprint summarized in Section 8.7. Section S3 documents how GHB was revised after a first run of Simulation 15, and three extensions that were tried and not adopted. Section S4 lists the 290 coded works. Every number in this supplement is generated by the scripts released with the paper (`prisma/` and `simulations/`) and read into the text as LaTeX macros.

S1 SYSTEMATIC SEARCH

S1.1 Sources and query strings

The search was run on 8 October 2026 for records published from 1 January 2010 to the search date. PubMed was queried through the NCBI E-utilities with every term restricted to title and abstract (`[tiab]`). medRxiv and the arXiv records indexed by Europe PMC were queried through the Europe PMC REST API (source `PPR`, field `TITLE_ABS`). Europe PMC indexes only the life-science categories of arXiv. A direct search of the arXiv API was attempted but abandoned because the API rejected the full query; computer-science preprints are therefore covered only when they were also published in an indexed venue. Every query string is stored verbatim in `prisma/data/search_log.json` together with the hit counts; the PubMed string was

```
("small data"[tiab] OR "small sample*"[tiab] OR "limited data"[tiab] OR "data scarcity"[tiab] OR
"few-shot"[tiab] OR "low-resource"[tiab] OR "rare disease*"[tiab] OR "label-efficient"[tiab] OR
"data-efficient"[tiab] OR federated[tiab] OR decentralized[tiab] OR decentralised[tiab] OR
"privacy-preserving"[tiab]) AND (health*[tiab] OR clinical[tiab] OR medic*[tiab] OR
hospital*[tiab] OR patient*[tiab]) AND ("machine learning"[tiab] OR "deep learning"[tiab] OR
"transfer learning"[tiab] OR "synthetic data"[tiab] OR "data augmentation"[tiab] OR "active
learning"[tiab] OR "semi-supervised"[tiab] OR "self-supervised"[tiab] OR "foundation
model*"[tiab] OR "hierarchical model*"[tiab]) AND 2010/01/01:2026/10/08[dp]
```

Scopus, Web of Science, IEEE Xplore and the ACM Digital Library were not searched, because they require a subscription or an API key. Exports from them (RIS, BibTeX or CSV) can be added with `prisma/import_exports.py`; they are then deduplicated and screened with the other records, and the flow diagram and every number are regenerated.

S1.2 Screening rules

Records were deduplicated by DOI and by normalized title, keeping the PubMed version when a preprint had also been published. Titles and abstracts were then screened by the rules of Table S1, applied in order. The rules operationalize criteria I1, I2 and X1 to X3 of Section 5: they keep every record in which a scarcity term of the query refers to scarce, fragmented or poorly labeled data and

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Table S1. Screening rules, applied in order; the first rule that applies decides.

Rule	Excludes records whose	*n*
A1	abstract is missing, or language is not English	46
A2	publication type is editorial, letter, comment, erratum, guideline, consensus statement, protocol, audio-visual material or historical article, or which are retracted (X1)	54
R1	only data-scarcity term refers to a resource-limited setting ("low-resource settings, countries, hospitals")	421
R2	only data-scarcity term is "small sample" in a sentence on limitations or future work	264
R3	only decentralization term refers to decentralized clinical trials or care delivery	14
R4	title is a reporting guideline, checklist, statement, position paper, policy or consensus (X1)	8
R5	title and abstract contain no learning method	72
	Eligible on title and abstract	5,204

a learning method is present, and they remove the frequent uses of the query terms that do not refer to data scarcity.

S1.3 Verification by a single reviewer

A single reviewer judged two random samples of 100 records each against criteria I1, I2 and X1 to X3, reading title and abstract; each sample drew 60 or 40 records from the included side and as many from the excluded side of the rules then in use. The first sample led to the removal of one rule, which excluded records whose scarcity term appears only in the author keywords (4 of its 8 false exclusions came from it). A rule added after the first sample, which excluded "limited data" in the sense of a gap in clinical knowledge, removed small-training-set studies in 3 of the 7 sampled records it excluded in the second sample and was also removed. Against all 200 judgments, the final rules have a precision of 63% and a negative predictive value of 94%; the sensitivity within the samples is 94%, but the samples are balanced by design, so this is not the sensitivity in the full set. A stricter rule, which required the scarcity term in the title or in a sentence that also mentions data, labels, training or a model, raised precision to 73% but lowered sensitivity to 77% and was not adopted. The judgments are released in `prisma/data/verify_judgments.json`.

S1.4 Recall against the coded corpus

A search is usually validated by its relative recall against a reference set of known relevant studies. Our reference set is the coded corpus: 201 of its works were published since 2010 and evaluated on health data, 144 of these are indexed in the searched sources (found in Europe PMC by DOI or title), and the protocol query retrieved 31 of them (22%). To test whether a better query would have found them, we split the 140 PubMed-indexed works at random into two halves and evaluated broadened variants of the query on PubMed (Table S2). Adding 39 terms to the scarcity block (sample size, multi-site, external validation, generalizability, shift, privacy, annotation and related terms) and 16 terms to the method block (artificial intelligence, neural networks, prediction models and related terms) multiplied the number of records by thirteen and raised recall to about one half. The literature on learning from scarce health data is not well delimited by vocabulary. A large part of it reports a method or a failure without naming scarcity at all, which is why the paper's synthesis rests on the snowballed corpus.

S1.5 Dictionary coder and field-level map

Each eligible record was coded from its title and abstract by dictionaries for the six provenance channels, the ten scarcity classes and the data modality (`prisma/classify.py`). The primary channel is the channel whose terms match most often, with title matches counted twice. On the

Table S2. Number of PubMed records and relative recall on two random halves of the PubMed-indexed corpus works, for the protocol query and for broadened variants.

Query	PubMed records	Recall, first half	Recall, second half
Protocol query (Section 5.4)	5,885	17%	27%
Scarcity block broadened	40,584	39%	44%
Method block broadened	11,178	21%	31%
Both broadened	75,983	51%	54%

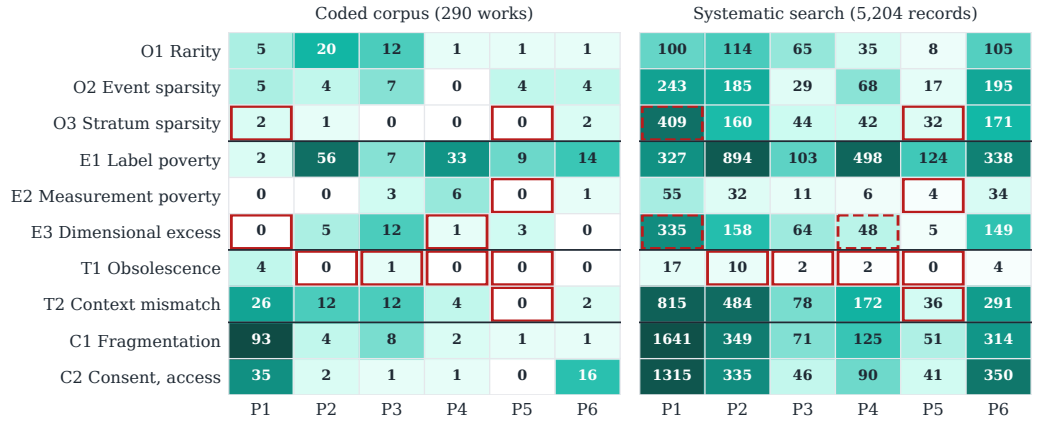


Fig. S1. Left: works of the coded corpus targeting each scarcity class through each channel, as in Figure 8 of the paper; red outlines mark the ten gap cells. Right: records of the systematic search coded by dictionary, shaded by percentile rank among the 60 cells. Solid outlines: gaps confirmed (bottom third of the field-level map); dashed outlines: gaps not confirmed.

147 corpus works whose abstracts are available in Europe PMC, the dictionary coder agreed with the manual primary channel in 72% of cases (Cohen's $\kappa = 0.65$) and recovered 63% of the manually assigned scarcity classes. Figure S1 sets the coverage of the coded corpus (Figure 8 of the paper) next to the field-level map. Because the dictionaries for stratum sparsity (which includes the term bias) and for dimensional excess (which includes omics terms) are broad, their rows are inflated in the field-level map; Section 5.4 of the paper discusses the three gap cells that the search does not confirm.

S2 SIMULATION STUDY OF THE BLUEPRINT

Section 8.7 of the paper summarizes these simulations; this section gives the full design and every figure. The simulations test the blueprint of Section 8 under its own model and under a logistic analogue. They check internal consistency and expose the cost of estimating the unknown quantities that Propositions 8.1 to 8.4 take as known; they are not evidence about real hospital networks. Unless stated otherwise, the network has 24 hospitals whose sizes are drawn once from a log-normal distribution with median 100 and clipped (16 to 479 patients), $\sigma^2 = 1$, $\delta = 10^{-5}$, clipping bound $C = 1$, and the privatized releases are averaged by 60 rounds of Metropolis-Hastings gossip on the ring with two-hop links of Figure 10. *Plug-in GHB* estimates everything it needs from the privatized releases and public quantities only: τ^2 by DerSimonian-Laird (Remark 8.3), the leave-one-out population estimate by inverse-variance weighting, and its error M from those

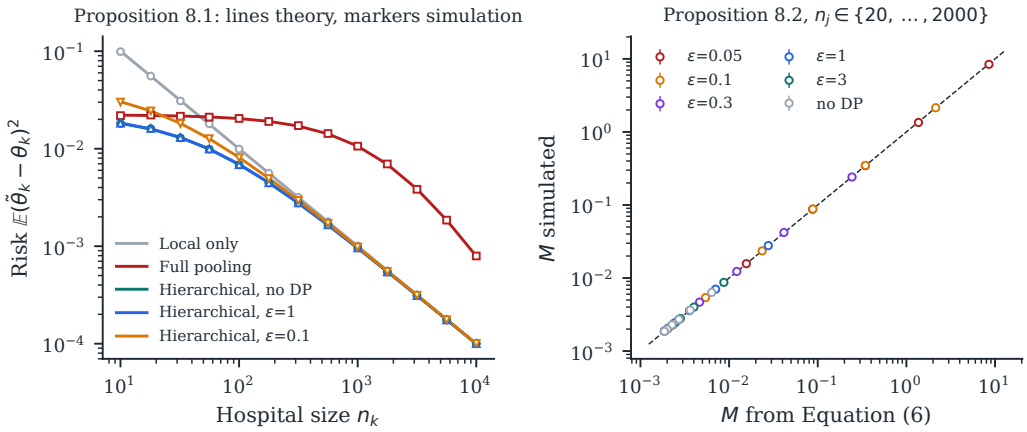


Fig. S2. Simulation 1. Left: risk against hospital size; lines are the closed forms of Figure 9, markers are simulated (±2 standard errors). Right: simulated M of the privatized leave-one-out mean against Equation (6), for 11 other hospitals of 20 to 2,000 patients and six privacy budgets.

weights. Risk is reported relative to local estimation, R_k/v_k with $v_k = \sigma^2/n_k$; values below 1 mean that a hospital gains. All code is in `simulations/`, and `run_all.py` regenerates every figure and number in about two minutes.

Simulation 1: the closed forms. Figure S2 repeats the setting of Figure 9 with 40,000 replicates per point. Across 101 comparisons of a simulated risk or M with its closed form, 98% lie within two Monte Carlo standard errors, and the largest relative deviation is 1.6%. Propositions 8.1 and 8.2 and the code that draws Figure 9 agree.

Simulation 2: heterogeneity. Figure S3 varies τ^2 from 10^{-4} to 1. Plug-in GHB is never worse than local estimation by more than Monte Carlo error: its worst hospital reaches a relative risk of 1.02 at most. A single global model (the sample-size weighted mean of all site estimates, computed without privacy) is the best choice when τ^2 is tiny, but its mean relative risk reaches 134 and its worst hospital 382 at $\tau^2 = 1$. Estimating τ^2 costs most when heterogeneity is smallest: at $\tau^2 = 10^{-4}$ and $\epsilon = 1$ the plug-in reaches 0.12 against 0.07 with the true τ^2. Gossip with 60 rounds is indistinguishable from exact averaging.

Simulation 3: who gains, and how many hospitals are needed. At $\tau^2 = 0.02$ and $\epsilon = 1$ (Figure S4, left), every hospital gains about the same number of virtual patients, $B \approx 46$, as Equation (5) predicts. The hospitals with 60 patients or fewer multiply their effective sample size by 2.28 on average, and those with 300 or more by 1.08; the smallest gain of any hospital is 1.05. The global model multiplies the effective sample size of small hospitals by 1.44 but cuts that of large hospitals to 0.15 of their own. With few hospitals, τ^2 is poorly estimated (Figure S4, right). With three hospitals the mean relative risk is 0.91 with DerSimonian-Laird against 0.78 with the true τ^2, and the gap closes as the network grows (0.68 against 0.66 with 48 hospitals). The posterior mean of τ^2 under a half-Cauchy(0.5) prior on τ, the weakly informative choice that Remark 8.3 recommends, does better than DerSimonian-Laird from four hospitals upward but not with three.

Simulation 4: gossip rounds. Figure S5 shows the mean relative risk of plug-in GHB after T rounds of gossip on the four topologies of Figure 10. The risk comes within 2% of its exact-average value

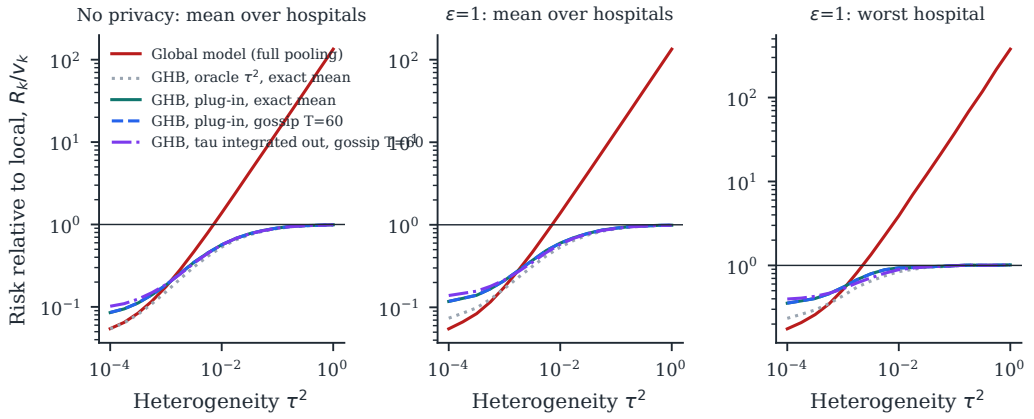


Fig. S3. Simulation 2. Risk relative to local estimation as a function of heterogeneity, mean over hospitals without privacy and with $\epsilon = 1$, and for the worst hospital with $\epsilon = 1$.

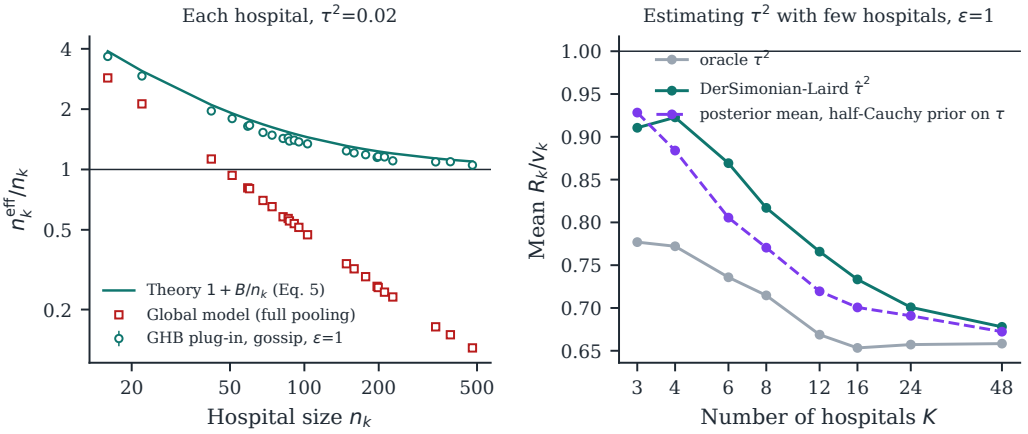


Fig. S4. Simulation 3. Left: effective sample size relative to own size for each hospital, plug-in GHB and a global model; the line is $1 + B/n_k$. Right: mean relative risk against the number of hospitals for three ways of obtaining τ^2.

after 35 rounds on the ring, 22 on the regional clusters, 11 on the sparse random graph and 14 on the ring with two-hop links, far fewer than the rounds that the worst-case bound of Proposition 8.4 requires for a consensus error of 1% (203, 127, 73 and 69). The bound is conservative for the risk: it controls the consensus error of the releases, while the risk depends on that error only through a population estimate that the shrinkage weight already discounts. After a single round on the ring, GHB is worse than local estimation (relative risk 1.13), so a hospital should not use the borrowed estimate before the gossip has mixed.

Simulation 5: joint release by small hospitals. Section 8.3 proposes that small hospitals release jointly through secure aggregation, so that the privacy noise is calibrated to the group. Figure S6 shows that the remedy matters only when every hospital is small. In a network of 16 hospitals of

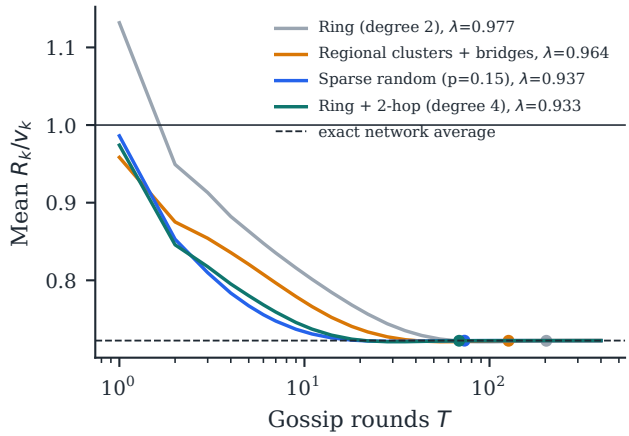


Fig. S5. Simulation 4. Mean relative risk of plug-in GHB against the number of gossip rounds ($\tau^2 = 0.02$, $\epsilon = 1$). Dots mark $\log(100)/(1-\lambda)$; the dashed line is the exact network average.

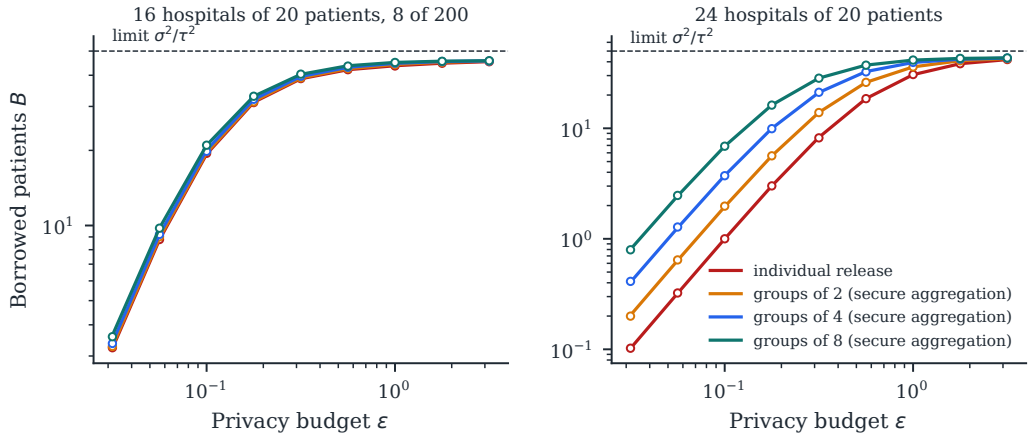


Fig. S6. Simulation 5. Borrowed patients B against the privacy budget for individual and joint release; lines are closed forms, markers simulation.

20 patients and 8 of 200, the large hospitals dominate the inverse-variance weighted population estimate, and groups of eight raise the borrowed patients only from 19.3 to 21.0 at $\epsilon = 0.1$. In a network of 24 hospitals of 20 patients, the same grouping raises B from 1.0 to 6.9 at $\epsilon = 0.1$ and from 30.6 to 41.4 at $\epsilon = 1$.

Simulation 6: structured heterogeneity. The 24 hospitals are split into four regions of six; θ_k is the sum of a regional effect with variance τ_r^2 and a hospital effect with variance $\tau_w^2 = 0.002$. The GMRF prior of Equation (7) links hospitals of the same region and is computed by Jacobi iterations on the privatized releases, followed by each hospital's local update with its own noise-free estimate. Its anchor term is switched off so that it uses the same information as the exchangeable prior, and its smoothing weight β was chosen on separate tuning replicates (the same value, $\beta \approx 46$, was selected

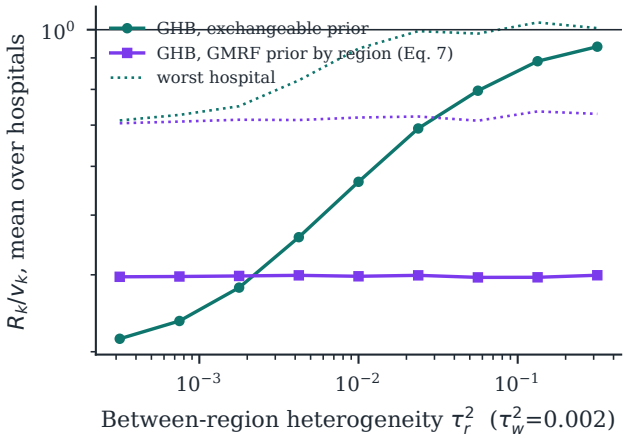


Fig. S7. Simulation 6. Mean relative risk (solid) and worst hospital (dotted) for the exchangeable and the regional GMRF prior, against the between-region variance ($\epsilon = 1$).

at every τ_r^2); the regions are assumed known. Figure S7 shows a crossover. When regions barely differ ($\tau_r^2 \lesssim 0.003$), the exchangeable prior is better (0.31 against 0.40 at the smallest τ_r^2), because the GMRF borrows only from five neighbors. When regional differences dominate, the exchangeable prior loses its gain (0.94) while the GMRF keeps it (0.40).

Simulation 7: logistic-regression heads. This simulation leaves the Gaussian model. Each hospital has its own logistic model on four standardized features and an intercept, $\theta_k = \mu + u_k$ with $u_k \sim \mathcal{N}(0, \tau^2 I)$, and features are clipped so that $\|x\| \le \sqrt{5}$. Here μ_0 is a published model (channel P3), equal to μ plus noise of standard deviation 0.3. Each hospital releases its mean score at μ_0, $g_k = \frac{1}{n_k}\sum_i x_i(y_i - \sigma(x_i^\top \mu_0))$, plus isotropic Gaussian noise. Its ℓ_2 sensitivity is $2\sqrt{5}/n_k$, and the noise is calibrated with zero-concentrated DP: $s_k = \Delta_k/\sqrt{2\rho}$ with $\rho + 2\sqrt{\rho \ln(1/\delta)} = \epsilon$. Under the public feature distribution, $\mathbb{E}g_k = m(\theta_k) - m(\mu_0)$ with $m(\theta) = \mathbb{E}[x\,\sigma(x^\top\theta)]$, so the releases are mapped to the parameter scale by moment matching, linearized first at the pooled moment estimate and then at each hospital's leave-one-out network estimate; their variances follow by the delta method. The anchor shapes that second point only for hospitals where it passes the conflict check against the anchor-free network estimate. GHB then borrows coordinatewise. Each hospital summarizes the posterior of τ (half-Cauchy(0.5) prior) by 12 quantiles, combines each component's network estimate with μ_0 by precision weighting unless it dropped the anchor, multiplies by the weakly informative default $\mathcal{N}(0, 2.5^2)$ [Gelman et al. 2008], and averages its maximum a posteriori fits under the components, weighted by the Laplace approximation of the marginal likelihood of its own data (Remark 8.3). The code is `simulations/ghb_logit.py`. The baselines are a local model with prior $\mathcal{N}(\mu_0, I)$, which uses the anchor but no other hospital, and a global model fitted on the pooled data of all hospitals without privacy. Table S3 and Figure S8 report the excess log-loss over the true model on 4,000 new patients per hospital, averaged over 100 replicates. At $\tau^2 = 0.05$ and $\epsilon = 1$, GHB reduces the excess log-loss of local estimation by 49% (0.0266 to 0.0135), below the global model trained without privacy (0.0173), and no hospital is worse off on average (largest ratio to local 0.85, against 2.48 for the global model). Without privacy GHB reaches 0.0106, better than the global model. At $\epsilon = 0.3$ it keeps most of its gain (0.0175 against 0.0273).

Table S3. Simulation 7. Excess log-loss over the true model (nats; lowest of the three methods in bold) and AUC, mean over 24 hospitals and 100 replicates. Oracle: the true model of each hospital. Global: one model on the pooled data, without privacy.

		Excess log-loss			AUC			
τ^2	ϵ	Local	Global	GHB	Local	Global	GHB	Oracle
0.01	1	0.0283	**0.0041**	0.0101	0.715	0.733	0.728	0.737
0.05	1	0.0266	0.0173	**0.0135**	0.723	0.728	0.731	0.743
0.2	1	0.0268	0.0601	**0.0214**	0.749	0.717	0.751	0.770
0.5	1	**0.0271**	0.1257	0.0285	0.784	0.688	0.781	0.804
0.05	0.3	0.0273	**0.0167**	0.0175	0.725	0.730	0.730	0.745
0.05	3	0.0271	0.0169	**0.0113**	0.722	0.728	0.733	0.743
0.05	10	0.0282	0.0170	**0.0109**	0.723	0.729	0.734	0.743
0.01	∞	0.0274	0.0042	**0.0042**	0.717	0.734	0.733	0.737
0.05	∞	0.0281	0.0170	**0.0106**	0.722	0.729	0.734	0.743
0.2	∞	0.0274	0.0599	**0.0199**	0.745	0.712	0.748	0.766
0.5	∞	0.0277	0.1259	**0.0260**	0.785	0.690	0.785	0.805

Development note (first version). Three components of an earlier version of this simulation were added after the first runs failed. The first version used the ridge releases without undoing the shrinkage, estimated τ^2 by DerSimonian-Laird and ignored μ_0. With privacy noise at $\epsilon = 1$ its estimates of τ^2 were erratic (median 0, mean several times the truth), and it was worse than local estimation at $\tau^2 = 0.2$ and at $\epsilon = 0.3$. Each component follows the design of the paper (post-processing, Remark 8.3 and layer L3), but they were chosen with knowledge of these failures, and the results should be read accordingly. A fourth change came from Simulation 11: the releases were first centred on the anchor ($c = \mu_0$), and the shrinkage was undone with the Fisher information at μ_0. When the anchor was badly wrong, this corrupted every release, and GHB without the anchor reached an excess log-loss of 0.34 against 0.06 for local estimation at an anchor error of two standard deviations. Centring on a fixed reference removes the dependence; Simulations 7, 8, 10 and 11 are reported with it. With anchor-centred releases, Simulation 7 had given a slightly smaller gain at $\epsilon = 1$ and no gain at $\epsilon = 0.3$. The first version of Simulation 6 also gave the GMRF prior the anchor μ_0, which the exchangeable prior did not have; the version reported here removes it.

Simulation 8: federated and meta-analytic baselines. The logistic world of Simulation 7, at $\epsilon = 1$ and four levels of heterogeneity. All private methods use the same releases, so they satisfy the same guarantee: one-shot federated averaging (the sample-size weighted mean of the releases, one model for every hospital), the random-effects meta-analytic mean (DerSimonian-Laird weights, one model for every hospital), and a Ditto-style personalized model, which fits each hospital's data with a proximal term $\frac{\lambda}{2}\|\theta - \bar{\theta}_{\text{FedAvg}}\|^2$. The weight λ is chosen from $\{0.3, 1, 3, 10, 30, 100\}$ on ten separate tuning replicates per setting, which favours Ditto-style. The global model fitted on the pooled data without privacy is a reference that no private method can match in general. Results (Figure S9, left; 40 replicates per setting): at $\tau^2 = 0.05$, GHB 0.0134, Ditto-style 0.0156, pooled 0.0167, meta-analysis 0.0303, federated averaging 0.0365, local 0.0274. At $\tau^2 = 0.5$, GHB 0.0282, Ditto-style 0.0265, local 0.0273. One-shot federated averaging is weak here because it averages noisy releases without personalization; multi-round federated training with differentially private gradients is compared in Simulation 15.

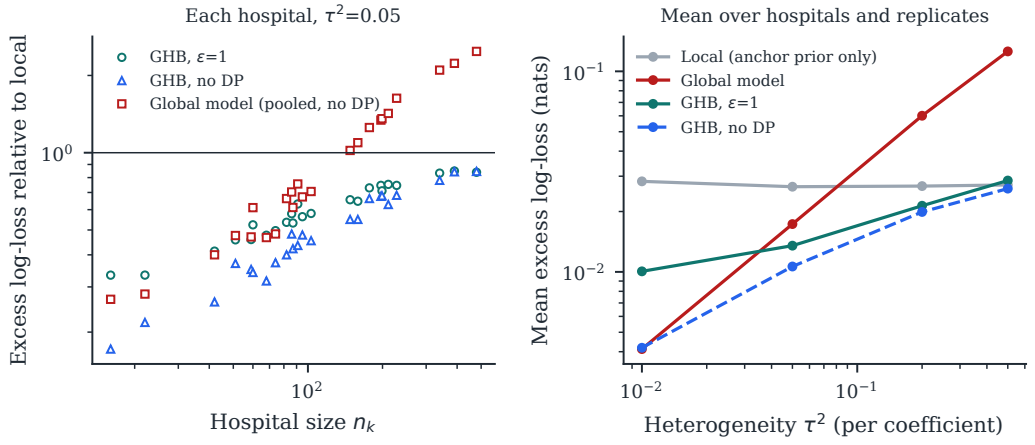


Fig. S8. Simulation 7. Left: excess log-loss relative to local estimation for each hospital ($\tau^2 = 0.05$). Right: mean excess log-loss against heterogeneity.

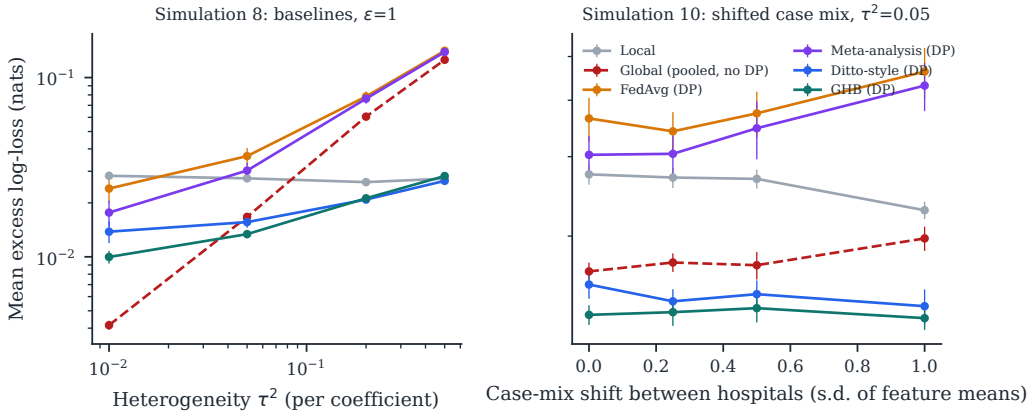


Fig. S9. Simulations 8 and 10. Mean excess log-loss of GHB and of private and non-private baselines against heterogeneity (left) and case-mix shift (right). Bars: ±2 standard errors.

Simulation 10: case-mix shift. As Simulation 8 at $\tau^2 = 0.05$, but each hospital draws its features from $\mathcal{N}(m_k, I)$ with $m_k \sim \mathcal{N}(0, s^2 I)$, while the de-shrinking still uses the public, unshifted feature distribution (Figure S9, right). At $s = 1$: GHB 0.0132, Ditto-style 0.0140, pooled 0.0198, local 0.0228.

Simulation 9: atypical hospitals. The network of Simulation 2 at $\tau^2 = 0.02$ and $\epsilon = 1$, with Gaussian site effects, Student-t_3 site effects of the same variance, or Gaussian effects plus one hospital at $u = 6\tau$, placed at the smallest or at the largest hospital. Three estimators: Gaussian GHB; robust GHB, which replaces the Gaussian prior $\mathcal{N}(\tilde{\mu}_k, \hat{\tau}^2 + M_k)$ by a Student-t_3 prior of the same variance (posterior mean by quadrature); and GHB with opt-out, where a hospital keeps its local estimate when $|\hat{\theta}_k - \tilde{\mu}_k| / \sqrt{v_k + \hat{\tau}^2 + M_k} > 3$. Both variants change only the hospital's own computation and cost no privacy. Mean relative risk: Gaussian effects 0.72 (robust 0.76), t_3 effects 0.69 (robust 0.65). The outlying hospital: 6.7 (robust 4.6, opt-out 4.4) when it is the smallest, 1.11 (robust 1.06) when it

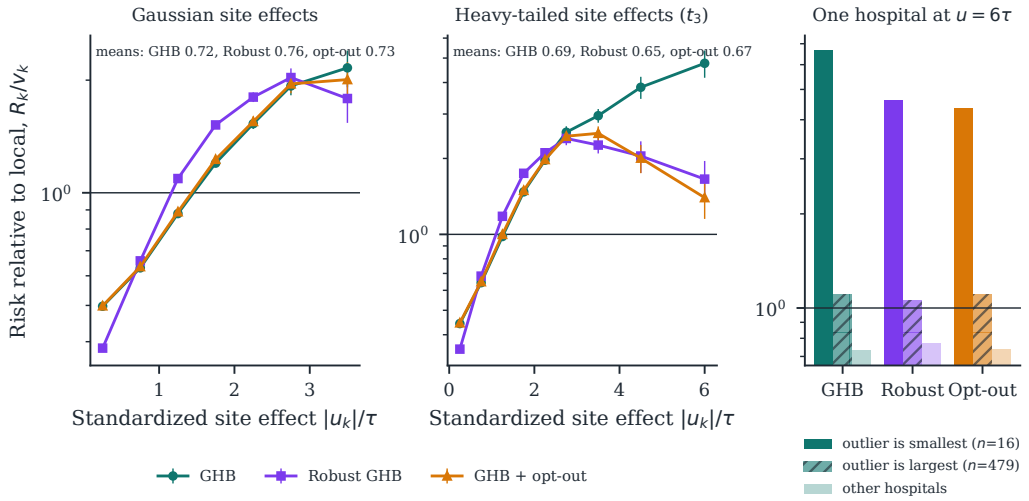


Fig. S10. Simulation 9. Left and centre: relative risk against the standardized site effect for Gaussian and heavy-tailed effects. Right: the outlying hospital and the others, when the outlier is the smallest or the largest hospital.

is the largest (Figure S10). With 16 patients, the standard error of the local estimate is as large as the deviation, so no test based on the hospital's own data can detect it reliably.

Simulation 11: a wrong anchor. The logistic world at $\tau^2 = 0.05$ and $\epsilon = 1$, with the anchor wrong by $\mathcal{N}(0, e^2 I)$ for e from 0 to 3. GHB uses the anchor with its uncertainty stated honestly (e), overconfidently (0.3 whatever e), overconfidently with a conflict check (the anchor is dropped when $|\mu_0 - \tilde{\mu}_k| > 3\sqrt{0.3^2 + M_k}$ in any coordinate), or not at all. The local model uses $\mathcal{N}(\mu_0, I)$ as its prior and degrades with the anchor (Figure S11). Honest uncertainty and the conflict check keep the excess log-loss at or below 0.0166 at every error, GHB without the anchor at or below 0.0167; an overconfident anchor two standard deviations off reaches 0.0352, and local estimation 0.0605.

Simulation 12: dishonest hospitals. The network of Simulation 2 at $\tau^2 = 0.02$ and $\epsilon = 1$. The one or three largest hospitals release $\mu + b\tau$ instead of their privatized estimate; sizes are public, so they cannot change their weights. Robust GHB obtains all releases by flooding, computes an inverse-variance weighted median and a MAD-based spread, drops releases more than three spreads away, and runs GHB on the rest; this is post-processing. Honest hospitals' mean relative risk (Figure S12): plain GHB never exceeds 1.00, because a large lie inflates $\hat{\tau}^2$ and switches borrowing off; robust GHB keeps 0.72 against large lies, but costs 0.81 against 0.73 when nobody lies, because the screen sometimes drops honest outlying releases.

Simulation 13: link failures. The network of Simulation 2 with every link failing independently in each round with probability p; the Metropolis-Hastings weights are recomputed on the working links, so each round's matrix stays symmetric and doubly stochastic. With $p = 0.8$, the mean relative risk is 0.77 after 20 rounds and 0.73 after 60, against 0.72 with exact averaging (Figure S13).

Simulation 14: repeated releases. For a total budget of $\epsilon = 1$, $\delta = 10^{-5}$ spread over r refits on the same patients, the borrowed patients B after the last refit (closed form, network of Simulation 2) under basic composition ($(\epsilon/r, \delta/r)$ per release) and under zero-concentrated accounting (the

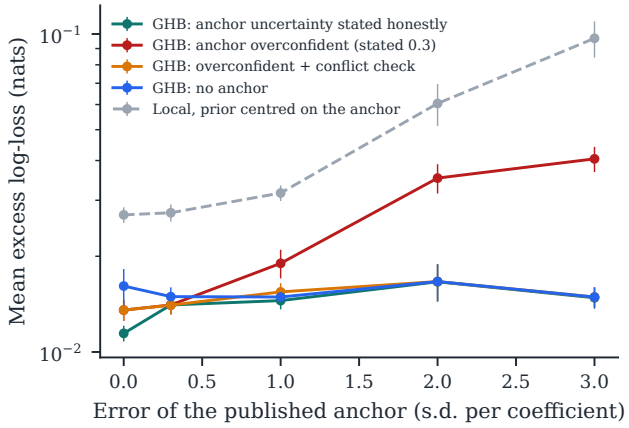


Fig. S11. Simulation 11. Mean excess log-loss against the error of the published anchor.

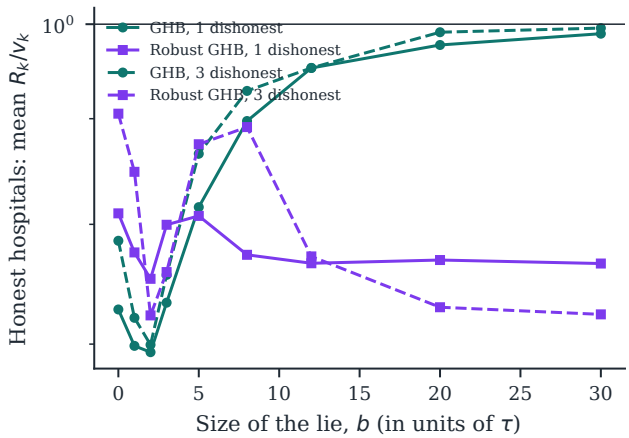


Fig. S12. Simulation 12. Honest hospitals' mean relative risk against the size of the lie, for one and three dishonest hospitals.

Gaussian mechanism with noise s is ρ-zCDP with $\rho = \Delta^2/(2s^2)$; costs add; ρ-zCDP implies $(\rho + 2\sqrt{\rho \ln(1/\delta)}, \delta)$-DP). After 64 refits, $B = 1.5$ under basic composition and 32.4 under zCDP, against 46.1 for a single release (Figure S14).

Simulation 15: iterative federated learning. The logistic world of Simulation 8 (24 hospitals, ring with two-hop links, $\delta = 10^{-5}$), GHB against multi-round training at the same privacy budget per hospital. GHB is the revised version of Section S3; the earlier version is reported alongside it. Per-patient gradients of the logistic loss are $(p - y)x$ with $\|x\| \le \sqrt{5}$, so they need no clipping, and replacing one patient moves a hospital's mean gradient by at most $2\sqrt{5}/n_k$. Every iterative method adds Gaussian noise to each hospital's mean gradient in every round and is accounted with zero-concentrated DP: with R rounds, each round spends ρ/R, and the total ρ satisfies $\rho + 2\sqrt{\rho \ln(1/\delta)} = \epsilon$.

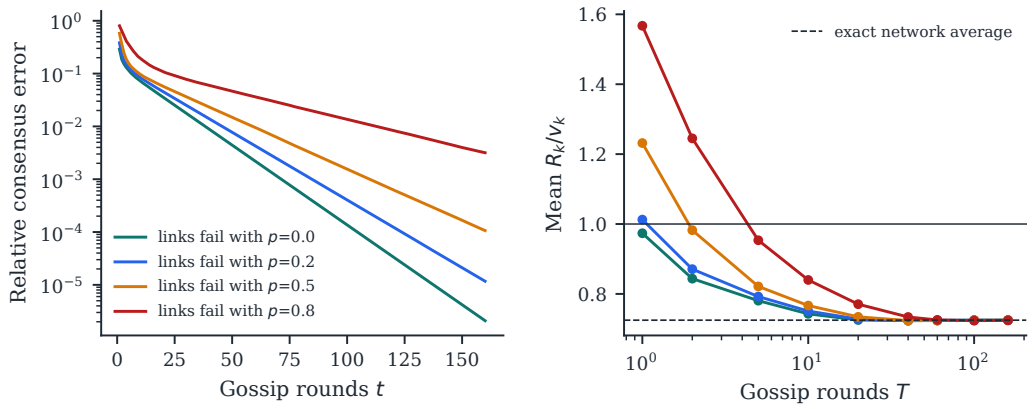


Fig. S13. Simulation 13. Left: consensus error against rounds. Right: mean relative risk of plug-in GHB against rounds, for four link-failure probabilities.

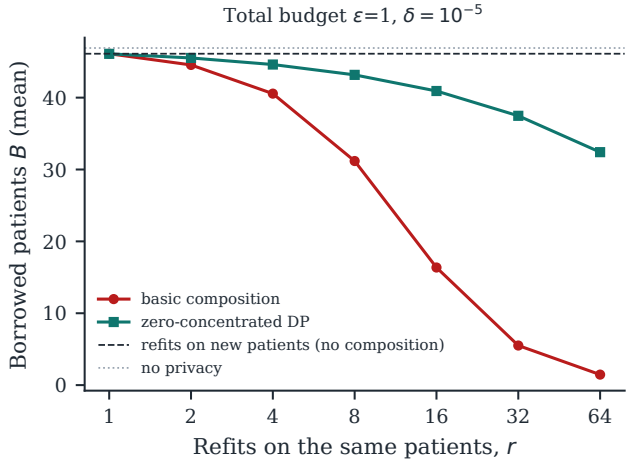


Fig. S14. Simulation 14. Borrowed patients after r refits on the same patients, for a fixed total budget.

GHB's single release is accounted in the same way. All methods start from, or shrink toward, the published anchor μ_0, which GHB also uses.

- *DP-FedAvg* [McMahan et al. 2017, 2018], full-batch: a server holds one model; in each round every hospital sends its noisy mean gradient at that model, and the server takes a step of size η along the sample-size weighted average.
- *DP-FedAvg with fine-tuning*: each hospital then takes E plain gradient steps (step size 1) on its own data, starting from the global model; this costs no privacy because the data stay local.
- *Decentralized DP-SGD* [Lian et al. 2017], the algorithm behind swarm learning [Warnat-Herresthal et al. 2021]: no server; each hospital steps along its own noisy gradient and averages models with its neighbours using the Metropolis-Hastings weights of GHB's graph (adapt, then combine), followed by optional fine-tuning.

- *Clustered DP-FedAvg*, in the style of IFCA [Ghosh et al. 2020]. IFCA lets each hospital choose the cluster whose model fits its data best in every round. Under DP this choice has sensitivity of order 1, not $1/n_k$, because one patient can move a hospital's whole gradient from one cluster to another. Hospitals therefore make one private ridge release with half the budget, the releases are clustered by weighted k-means (post-processing), and DP-FedAvg runs within each cluster with the other half, followed by optional fine-tuning.
- *Local gradient descent from the anchor*, a control that uses no other hospital: E gradient steps on the hospital's own data from μ_0, with early stopping as the only regularizer.
- *Ditto-style* as in Simulation 8, and *GHB with the true* τ^2 (the marginal heterogeneity, regional plus within-region in the regional world), a reference that no method can use.

Hyperparameters are chosen on 20 separate replicates per setting by the mean excess log-loss, which requires access to the true models: rounds $R \in \{1, 2, 3, 5, 10, 30, 100\}$ ($\{3, 10, 30, 100\}$ for decentralized training), step size $\eta \in \{0.2, 0.5, 2\}$, fine-tuning steps $E \in \{0, 1, 2, 3, 5, 10, 30, 100\}$ and clusters $C \in \{2, 3, 4\}$. Table S5 lists the choices. The only choices at the edge of a grid are $R = 1$, the smallest meaningful number of rounds, and $\eta = 0.2$ for decentralized training, where smaller steps would approach the anchor-only control. GHB has no tuning parameter. Settings: $\tau^2 \in \{0.01, 0.05, 0.2, 0.5\}$ at $\epsilon = 1$; $\epsilon \in \{0.3, \infty\}$ at $\tau^2 = 0.05$; a regional world in which the 24 hospitals form four regions, with between-region variance 0.2 and within-region variance 0.02 per coefficient, where clustering should help; an anchor wrong by $\mathcal{N}(0, I)$ instead of $\mathcal{N}(0, 0.3^2 I)$; $\tau^2 = 0.1$ at $\epsilon = 0.5$; and a network of 48 hospitals (sizes drawn as for the 24, ring with two-hop links). The last four settings (marked $*$) were not used while revising GHB; the development runs used other seeds, an anchor error of two standard deviations and a three-region world. Results are means over 60 further replicates (Table S4, Figure S15).

GHB is the most accurate private method in six of the ten settings ($\tau^2 = 0.01$ and 0.05, $\epsilon = 0.3$, and the held-out regional, $\tau^2 = 0.1/\epsilon = 0.5$ and 48-hospital settings). The tuned Ditto-style model, which shrinks toward the average of GHB's own releases, is slightly more accurate at $\tau^2 = 0.2$, without privacy and with a wrong anchor; at $\tau^2 = 0.5$ no private method beats local estimation. GHB is more accurate than DP-FedAvg with fine-tuning, decentralized DP-SGD and clustered training in every private setting. At $\epsilon = 0.3$, the local control that starts from the anchor reaches 0.0205 against 0.0175 for GHB, and with a wrong anchor it becomes worse than local estimation (0.0419 against 0.0346). GHB's worst hospital is at most 1.06 times its local error in every private setting except $\tau^2 = 0.5$ (1.15), whereas fine-tuning's worst hospital is 1.39 to 2.96 times worse than alone; the hospitals that lose are the large ones. GHB with the true τ^2 remains better still (0.0158 at $\epsilon = 0.3$), so estimating heterogeneity from noisy releases is what is left to improve. The earlier version of GHB reached 0.0259 at $\epsilon = 0.3$ and 0.0166 at $\tau^2 = 0.05$.

S3 REVISION OF GHB AND EXTENSIONS TRIED

The first run of Simulation 15 showed that the earlier GHB lost to DP-FedAvg with fine-tuning when hospitals were nearly identical and when the budget was strict. A diagnosis on development seeds (900 to 904, used by no reported simulation) found two causes. First, its plug-in τ^2, the posterior mean under a half-Cauchy prior, is driven by the heavy prior tail when releases are noisy: it averaged 0.8 to 1.1 against a true 0.05 at $\epsilon = 0.3$, so GHB barely borrowed. Second, its ridge releases ($\lambda = 0.2$) have five times the sensitivity of a score, and undoing the ridge to first order at a fixed reference left a bias in the releases of the largest hospitals without privacy (their squared errors were 2.5 to 9 times their stated variances). The revision makes three changes, each following the design of Section 8: a score release at the anchor, inverted by public moment matching at anchor-free or check-gated points (scope (d) of Section 8.3); τ integrated out over its posterior,

Table S4. Simulation 15. Mean excess log-loss (nats) over the hospitals and 60 replicates; lowest of the methods available to a hospital in bold (the earlier version of GHB, the oracle and the pooled model are references). FT: fine-tuning. $*$: setting not used while revising GHB.

Setting	Local	Anchor GD	GHB	Ditto-style	DP-FedAvg	DP-FedAvg + FT	Decentr. DP-SGD	Clustered DP-FedAvg	GHB, earlier	GHB, true τ^2	Pooled, no DP
$\tau^2 = 0.01$	0.0284	0.0166	**0.0101**	0.0122	0.0156	0.0114	0.0166	0.0163	0.0155	0.0081	0.0043
$\tau^2 = 0.05$	0.0271	0.0190	**0.0138**	0.0154	0.0274	0.0153	0.0191	0.0214	0.0166	0.0124	0.0173
$\tau^2 = 0.2$	0.0267	0.0244	0.0217	**0.0205**	0.0732	0.0232	0.0244	0.0244	0.0207	0.0194	0.0610
$\tau^2 = 0.5$	**0.0260**	0.0293	0.0275	0.0260	0.1366	0.0287	0.0294	0.0294	0.0273	0.0240	0.1244
$\epsilon = 0.3$	0.0269	0.0205	**0.0175**	0.0306	0.0452	0.0196	0.0248	0.0206	0.0259	0.0158	0.0170
no DP	0.0278	0.0190	0.0104	**0.0101**	0.0167	0.0113	0.0118	0.0138	0.0130	0.0097	0.0167
regional*	0.0287	0.0265	**0.0209**	0.0212	0.0631	0.0249	0.0264	0.0271	0.0219	0.0199	0.0485
wrong anchor*	0.0346	0.0419	0.0162	**0.0148**	0.0446	0.0210	0.0418	0.0463	0.0194	0.0150	0.0169
$\tau^2 = 0.1$, $\epsilon = 0.5$*	0.0269	0.0228	**0.0184**	0.0238	0.0526	0.0217	0.0239	0.0229	0.0227	0.0171	0.0327
$K = 48$*	0.0324	0.0224	**0.0144**	0.0146	0.0258	0.0160	0.0225	0.0241	0.0170	0.0132	0.0172

Table S5. Simulation 15. Hyperparameters chosen on the tuning replicates: rounds R, step size η, fine-tuning steps E, clusters C, and the proximal weight λ of the Ditto-style model.

Setting	DP-FedAvg + FT	Decentralized DP-SGD	Clustered DP-FedAvg	Anchor GD	Ditto λ
$\tau^2 = 0.01$	R=3, η=2.0, E=2	R=3, η=0.2, E=5	C=2, R=1, η=0.5, E=5	E=5	10
$\tau^2 = 0.05$	R=10, η=0.5, E=5	R=3, η=0.2, E=5	C=2, R=5, η=0.2, E=5	E=5	10
$\tau^2 = 0.2$	R=2, η=2.0, E=10	R=3, η=0.2, E=10	C=3, R=1, η=0.5, E=10	E=10	3
$\tau^2 = 0.5$	R=30, η=0.2, E=10	R=3, η=0.2, E=10	C=2, R=1, η=0.5, E=10	E=10	1
$\epsilon = 0.3$	R=3, η=0.5, E=5	R=3, η=0.2, E=10	C=2, R=1, η=0.2, E=5	E=5	1
no DP	R=100, η=2.0, E=3	R=100, η=0.2, E=3	C=2, R=30, η=0.5, E=3	E=5	30
regional*	R=30, η=0.5, E=10	R=3, η=0.2, E=10	C=2, R=5, η=0.2, E=10	E=10	3
wrong anchor*	R=100, η=0.2, E=5	R=30, η=0.2, E=10	C=4, R=1, η=2.0, E=30	E=30	10
$\tau^2 = 0.1$, $\epsilon = 0.5$*	R=5, η=0.5, E=5	R=3, η=0.2, E=10	C=2, R=2, η=0.2, E=10	E=10	3
$K = 48$*	R=3, η=2.0, E=3	R=3, η=0.2, E=5	C=2, R=10, η=0.2, E=5	E=5	10

Table S6. Development runs (20 replicates each, development seeds): mean excess log-loss of the earlier GHB, the revised GHB, and local estimation. Settings use the logistic world of Simulation 7; "shift" is a case-mix shift of one standard deviation, "anchor" an anchor wrong by two standard deviations.

	$\tau^2 = 0.05$			$\tau^2 = 0.5$	$\tau^2 = 0.05$, $\epsilon = 1$	
	$\epsilon = 0.3$	$\epsilon = 1$	no DP	$\epsilon = 1$	shift	anchor
Local	0.0261	0.0289	0.0275	0.0272	0.0239	0.0581
GHB, earlier version	0.0243	0.0166	0.0134	0.0268	0.0151	0.0194
GHB, revised	0.0158	0.0143	0.0106	0.0269	0.0138	0.0160

weighted by each hospital's own marginal likelihood (Remark 8.3); and the anchor conflict check by default, decided once per hospital on anchor-free information. Table S6 gives the development runs that chose between the alternatives; the scripts are in `simulations/dev/`.

Alternatives rejected on the development runs. Releasing $X^\top y/n$ instead of the score: calibrated, but about twice the sampling variance for the slopes, because it ignores the hospital's own features. Ridge releases centred on the anchor ($\lambda = 1$ or 0.2) or on zero: competitive when the anchor is right, but two to three times worse with a wrong anchor (0.032 to 0.045 against 0.016). Plug-in alternatives to integrating τ out: the posterior median or the posterior mean of the precision borrowed more at small τ^2 but became worse than local estimation at $\tau^2 = 0.5$ (up to 0.041 against 0.028). An additional Student-t tail on the integrated prior changed nothing beyond Monte Carlo error. Without a weakly informative default prior, the widest components left tiny hospitals' fits unregularized and logistic

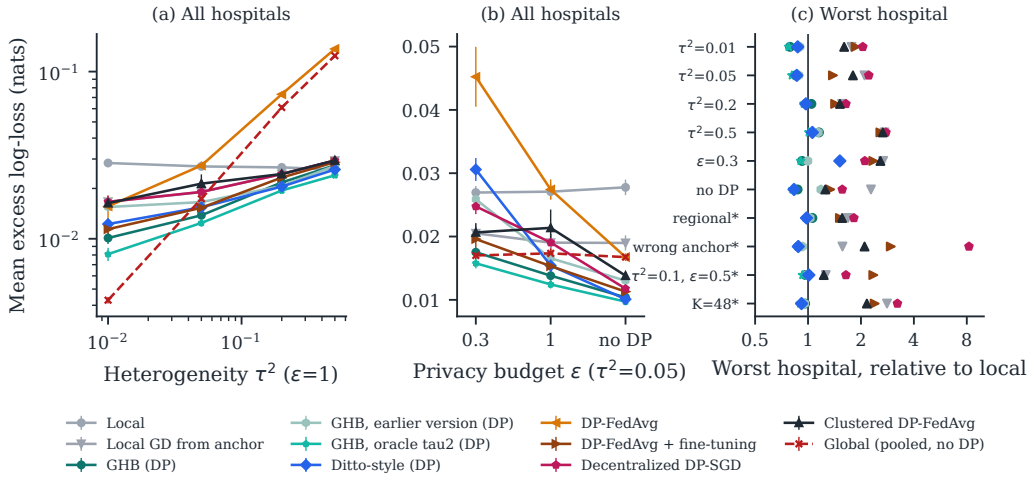


Fig. S15. Simulation 15. (a) Mean excess log-loss against heterogeneity at $\epsilon = 1$. (b) Against the privacy budget at $\tau^2 = 0.05$. (c) The worst hospital's error relative to its own local estimate; right of 1, some hospital is worse off than alone.

separation made some diverge; with a conflict check applied separately per component, a wrong anchor re-entered through the wide components.

Extensions tried and not adopted. Three extensions proposed in Section 8.3 and the agenda were developed in the same way (`dev/ext.py`, `ext2.py`, `ext3.py`). (1) *Joint release* by groups of g consecutive hospitals through secure aggregation, each group entering borrowing as one unit whose heterogeneity is diluted by its size weights: groups of three over all hospitals lowered the error by at most 12% (0.0140 against 0.0160 at $\tau^2 = 0.01$, $\epsilon = 0.3$; 0.0125 against 0.0133 at $\tau^2 = 0.05$, $\epsilon = 1$) and never raised it beyond Monte Carlo error. It is worth its cost only where secure aggregation is available anyway. (2) *Two release rounds* under zCDP, the second score evaluated at each hospital's public network estimate: no gain in any setting (for example 0.0176 against 0.0177 at $\epsilon = 0.3$, 0.0138 against 0.0135 at $\epsilon = 1$), because splitting the budget cancels the better linearization point. (3) *A two-level prior* for public region labels (hospital within region within network), with both variance components integrated out: a small gain in a three-region world at $\epsilon = 1$ (0.0158 against 0.0164) but losses elsewhere (0.0158 against 0.0141 without regions, 0.0215 against 0.0175 at $\epsilon = 0.3$): two variance components cannot be estimated reliably from 24 noisy releases.

S4 THE CODED CORPUS

Table S7 lists all 290 works of the corpus with their coding under the scheme of Table 3 of the paper. The same data are released as a CSV file, together with the scripts that compute every count in Sections 5 and 7 and draw Figures 7 and 8.

Table S7. The coded corpus (290 works), grouped by primary provenance channel and sorted by year. Sec.: secondary channel. Scarcity: classes of Figure 3 the work targets or documents. Type: Method, Theory, Study (single-dataset empirical study), Multi-site (training or validation across institutions), Resource (dataset, benchmark or infrastructure), Review, Evidence (documented failure or risk). General: method not specific to health data.

Work	Modality	Sec.	Scarcity	Type
P1 Pooling (102)				
DerSimonian and Laird [1986]	Tabular	P3	C1, T2	Method
Sweeney [2002]	Tabular		C2	Method
Xiao and Boyd [2004]	General		C1	Theory
Boyd et al. [2006]	General		C1	Theory
Dwork et al. [2006]	General		C2	Theory
Gentry [2009]	General		C2	Theory
Nedić and Ozdaglar [2009]	General		C1	Theory
Riley et al. [2010]	Tabular		C1, O2	Review
Overhage et al. [2012]	EHR		C2	Resource
Wu et al. [2012]	Tabular		C1	Method
Jiang et al. [2013]	Tabular		C1	Resource
Gaye et al. [2014]	Tabular		C1, C2	Resource
Hripcsak et al. [2015]	EHR		C1, C2	Resource
Lu et al. [2015]	Tabular		C1	Resource
Abadi et al. [2016]	General		C2	Method
Jochems et al. [2016]	Tabular		C1	Multi-site
Johnson et al. [2016]	EHR		C1, C2	Resource
Mandel et al. [2016]	EHR		C2	Resource
Blanchard et al. [2017]	General		C1	Method
Bonawitz et al. [2017]	General		C1, C2	Method
Jochems et al. [2017]	Tabular		C1	Multi-site
Kim et al. [2017]	EHR		C1	Method
Lian et al. [2017]	General		C1	Theory
McMahan et al. [2017]	General		C1	Method
Mironov [2017]	General		C2	Theory
Papernot et al. [2017]	General	P4	C2	Method
Smith et al. [2017]	General	P3	C1, T2	Method
Vanhaesebrouck et al. [2017]	General	P3	C1, T2	Method
Bellet et al. [2018]	General	P3	C1, T2	Method
Brisimi et al. [2018]	EHR		C1	Method
Chang et al. [2018]	Imaging		C1	Multi-site
Cho et al. [2018]	Omics		C1, C2	Method
Gupta and Raskar [2018]	General		C1	Method
Lee et al. [2018]	EHR		C1, C2	Method
McMahan et al. [2018]	General		C1, C2	Method
Reps et al. [2018]	EHR		C2	Resource
Vepakomma et al. [2018]	Multimodal		C1, C2	Method
Zhao et al. [2018]	General		C1, T2	Method
Choudhury et al. [2019]	EHR		C1, C2	Method
Huang et al. [2019]	EHR		C1, T2	Multi-site
Jordan et al. [2019]	General		C1	Theory
Li et al. [2019]	Imaging		C1, C2	Method
Mohri et al. [2019]	General		C1, O3	Method
Poirot et al. [2019]	Imaging		C1, C2	Method
Raisaro et al. [2019]	Multimodal		C1, C2	Method
Roy et al. [2019]	Imaging		C1	Method
Sheller et al. [2019]	Imaging		C1	Method
Suchard et al. [2019]	EHR		C1, C2, O2	Multi-site
Yang et al. [2019]	General		C1	Review
Beyan et al. [2020]	Multimodal		C1, C2	Resource
Chen et al. [2020b]	Signals	P2	C1, T2	Method
Deist et al. [2020]	Tabular		C1	Multi-site
T. Dinh et al. [2020]	General	P3	C1, T2	Method
Duan et al. [2020b]	EHR		C1, O2	Method
Duan et al. [2020a]	EHR		C1, O2	Method
Fallah et al. [2020]	General	P2	C1, T2	Method
Ghosh et al. [2020]	General		C1, T2	Method
Hsieh et al. [2020]	General		C1, T2	Method
Kaissis et al. [2020]	Imaging		C1, C2	Review
Karimireddy et al. [2020]	General		C1, T2	Method
Koloskova et al. [2020]	General		C1	Theory
Li et al. [2020c]	General		C1, T2	Method
Li et al. [2020a]	Imaging	P2	C1, T2	Multi-site

Work	Modality	Sec.	Scarcity	Type
Lin et al. [2020]	General	P4	C1, T2	Method
Rieke et al. [2020]	Multimodal		C1, C2	Review
Roth et al. [2020]	Imaging		C1, T2	Multi-site
Sheller et al. [2020]	Imaging		C1, O1	Multi-site
Wei et al. [2020]	General		C1, C2	Method
Bai et al. [2021]	Imaging		C1, T1	Multi-site
Collins et al. [2021]	General	P2	C1, T2	Method
Dayan et al. [2021]	Multimodal		C1, T1, O2	Multi-site
Dou et al. [2021]	Imaging		C1, T1	Multi-site
Froelicher et al. [2021]	Omics		C1, C2	Method
Guo et al. [2021]	Imaging		C1	Method
Huang et al. [2021a]	General		C1, T2	Method
Kairouz et al. [2021]	General		C1	Review
Kaissis et al. [2021]	Imaging		C1, C2	Method
Li et al. [2021a]	General	P3	C1, T2	Method
Li et al. [2021b]	General		C1, T2	Method
Liu et al. [2021]	Imaging	P6	C1, T2	Method
Marfoq et al. [2021]	General		C1, T2	Method
Sarma et al. [2021]	Imaging		C1	Multi-site
Tramèr and Boneh [2021]	General	P2	C2	Method
Vaid et al. [2021]	EHR		C1, T1	Multi-site
Warnat-Herresthal et al. [2021]	Multimodal		C1, O3	Multi-site
Xu et al. [2021]	Multimodal		C1	Review
Ziller et al. [2021]	Imaging		C2	Method
Adnan et al. [2022]	Imaging		C1, C2	Method
Cyffers and Bellet [2022]	General		C1, C2	Theory
Jiang et al. [2022]	Imaging		C1, T2	Method
Lu et al. [2022]	Imaging		C1, E1	Method
Nguyen et al. [2022]	Multimodal		C1	Review
Ogier du Terrail et al. [2022]	Multimodal		C1, T2	Resource
Pati et al. [2022]	Imaging		C1, O1	Multi-site
Saldanha et al. [2022]	Imaging		C1	Multi-site
Tedeschini et al. [2022]	Imaging		C1	Method
Xu et al. [2022]	Imaging		C1, T2	Method
Kalra et al. [2023]	Imaging		C1, C2	Method
Lu et al. [2023]	General		C1	Method
Ogier du Terrail et al. [2023]	Imaging		C1, O1	Multi-site
Tan et al. [2023]	General		C1, T2	Review
Yan et al. [2023]	Imaging	P4	C1, E1	Method
P2 Proxy (60)				
Baxter [2000]	General		O1, E1	Theory
Ben-David et al. [2010]	General		T2	Theory
Pan and Yang [2010]	General		E1, T2	Review
Choi et al. [2016]	EHR		E1, T2	Method
Gulshan et al. [2016]	Imaging		E1	Study
Maurer et al. [2016]	General		O1	Theory
Shin et al. [2016]	Imaging		E1	Study
Tajbakhsh et al. [2016]	Imaging		E1	Study
Esteva et al. [2017]	Imaging		E1	Study
Finn et al. [2017]	General		O1, E1	Method
Kamnitsas et al. [2017]	Imaging	P4	T2	Method
Snell et al. [2017]	General		O1, E1	Method
Alsentzer et al. [2019]	Text		E1	Method
Chen et al. [2019]	Imaging		E1	Method
Gurovich et al. [2019]	Imaging		O1	Method
Prabhu et al. [2019]	Imaging		O1, O3	Method
Raghu et al. [2019]	Imaging		E1	Study
Zhang et al. [2019]	EHR		O1, E1	Method
Brown et al. [2020]	General		E1	Method
Lee et al. [2020]	Text		E1	Method
Li et al. [2020b]	EHR		E1, E3	Method
Mahajan et al. [2020]	Imaging		O1	Method
Roy et al. [2020]	Imaging		E1, O1	Method
Tripuraneni et al. [2020]	General		E1	Theory
Wang et al. [2020]	General		O1, E1	Review
Azizi et al. [2021]	Imaging	P4	E1, T2	Method
Du et al. [2021]	General		E1	Theory
Huang et al. [2021b]	Imaging	P4	E1	Method
Radford et al. [2021]	General		E1	Method
Rasmy et al. [2021]	EHR		E1, O2	Method
Sowrirajan et al. [2021]	Imaging	P4	E1	Method
Steinberg et al. [2021]	EHR		E1	Method

Work	Modality	Sec.	Scarcity	Type
Zhou et al. [2021]	Imaging	P4	E1	Method
Agrawal et al. [2022]	Text		E1	Method
Guan and Liu [2022]	Imaging		T2	Review
Hospedales et al. [2022]	General		O1, E1	Review
Hsieh et al. [2022]	Imaging		O1	Method
Hu et al. [2022]	General		E1	Method
Mei et al. [2022]	Imaging		E1	Resource
Tiu et al. [2022]	Imaging	P4	E1	Method
Wang et al. [2022]	Imaging	P4	E1	Method
Zhang et al. [2022]	Imaging	P4	E1	Method
Banerjee et al. [2023]	Multimodal	P3	O1	Review
Hegselmann et al. [2023]	Tabular	P3	E1, O2	Method
Hollmann et al. [2023]	Tabular	P3	E3, O2	Method
Jiang et al. [2023]	Text		E1	Method
Moor et al. [2023]	Multimodal		E1, O1	Review
Nori et al. [2023]	Text		E1	Study
Singhal et al. [2023]	Text		E1	Method
Theodoris et al. [2023]	Omics		E1, E3, O1	Method
Wornow et al. [2023]	EHR		E1, T2	Review
Zhou et al. [2023]	Imaging	P4	E1, T2	Method
Chen et al. [2024]	Imaging		E1, O1	Method
Cui et al. [2024]	Omics		E1, E3	Method
Lu et al. [2024]	Imaging		E1, O1	Method
Ma et al. [2024]	Imaging		E1	Method
Pai et al. [2024]	Imaging		E1, O1	Method
Vorontsov et al. [2024]	Imaging		E1, O1	Method
Xu et al. [2024]	Imaging		E1	Method
Hollmann et al. [2025]	Tabular	P3	E3, O2	Method
P3 Prior (25)				
Stein [1956]	General	P1	O1, C1	Theory
Efron and Morris [1973]	General	P1	O1, C1	Theory
Tibshirani [1996]	General		E3	Method
Steyerberg et al. [2000]	Tabular		O2, E3	Study
Lake et al. [2015]	General		O1	Method
Choi et al. [2017a]	EHR		E3, O1	Method
Rotmensch et al. [2017]	EHR	P4	E3	Method
Ma et al. [2018]	EHR		E3, O1	Method
Arjovsky et al. [2019]	General		T1, T2	Method
Raissi et al. [2019]	General		O1, E3	Method
Shang et al. [2019]	EHR		E3	Method
Subbaswamy et al. [2019]	EHR		T2	Method
Castro et al. [2020]	Imaging		T2, E2	Review
Kissas et al. [2020]	Imaging		E1	Method
Lagergren et al. [2020]	Tabular		E1, E3	Method
Sahli Costabal et al. [2020]	Signals		O1, E1	Method
Van Calster et al. [2020]	Tabular		O2	Study
Gulrajani and Lopez-Paz [2021]	General		T2	Study
Karniadakis et al. [2021]	General		O1, E3	Review
Köhler et al. [2021]	Text		O1	Resource
Lu et al. [2021a]	Tabular		O1	Method
Riley et al. [2021]	Tabular		O2	Study
Schölkopf et al. [2021]	General		T2	Review
Zhang et al. [2021]	Multimodal		T2	Study
von Rueden et al. [2023]	General		O1, E3	Review
P4 Unlabeled (23)				
Schölkopf et al. [2012]	General		E1	Theory
Agarwal et al. [2016]	EHR		E1, E2	Method
Halpern et al. [2016]	EHR	P3	E1, E2	Method
Bai et al. [2017]	Imaging		E1	Method
Ratner et al. [2017]	General	P3	E1, E2	Method
Wang et al. [2017]	Imaging		E1, E2	Resource
Ilse et al. [2018]	Imaging		E1	Method
Campanella et al. [2019]	Imaging		E1	Method
Cheplygina et al. [2019]	Imaging	P2	E1	Review
Fries et al. [2019]	Imaging		E1	Method
Irvin et al. [2019]	Imaging		E1, E2	Resource
Wang et al. [2019]	Text	P3	E1	Method
Yu et al. [2019]	Imaging		E1	Method
Chaitanya et al. [2020]	Imaging	P2	E1	Method
Chen et al. [2020a]	General	P2	E1	Method
Ouyang et al. [2020]	Imaging	P2	E1, O1	Method
Sohn et al. [2020]	General		E1	Method

Work	Modality	Sec.	Scarcity	Type
Tajbakhsh et al. [2020]	Imaging	P6	E1, E2	Review
Taleb et al. [2020]	Imaging	P2	E1	Method
Li et al. [2021c]	Imaging		E1	Method
Lu et al. [2021b]	Imaging		E1	Method
He et al. [2022]	General	P2	E1	Method
Krishnan et al. [2022]	Multimodal	P2	E1	Review
P5 Procurement (15)				
Peduzzi et al. [1996]	Tabular		O2	Theory
Settles [2009]	General		E1	Review
Figueroa et al. [2012]	Text		E1	Method
Cho et al. [2015]	Imaging		E1	Study
Gal et al. [2017]	General		E1	Method
Yang et al. [2017]	Imaging		E1	Method
Smailagic et al. [2018]	Imaging		E1	Method
Balki et al. [2019]	Imaging		E1, O1	Review
Ghorbani and Zou [2019]	General		C1	Method
Riley et al. [2019b]	Tabular		O2, E3	Theory
Riley et al. [2019a]	Tabular		E3	Theory
van Smeden et al. [2019]	Tabular	P3	O2	Theory
Riley et al. [2020]	Tabular		O2, E3	Theory
Budd et al. [2021]	Imaging		E1	Review
Nath et al. [2021]	Imaging		E1	Method
P6 Re-expression (32)				
Chawla et al. [2002]	Tabular		O2	Method
Goodfellow et al. [2014]	General		E1	Method
Choi et al. [2017b]	EHR		C2	Method
Bowles et al. [2018]	Imaging		E1	Method
Fernández et al. [2018]	Tabular		O2	Review
Frid-Adar et al. [2018]	Imaging		E1, O2	Study
Shin et al. [2018]	Imaging		E1, C2	Method
Walonoski et al. [2018]	EHR	P3	C2	Resource
Zhang et al. [2018]	General		E1	Method
Baowaly et al. [2019]	EHR		C2	Method
Beaulieu-Jones et al. [2019]	Tabular	P1	C2	Method
Jordon et al. [2019]	Tabular	P1	C2	Method
Shorten and Khoshgoftaar [2019]	General		E1	Review
Xu et al. [2019]	Tabular		C2	Method
Yi et al. [2019]	Imaging		E1	Review
Zhao et al. [2019]	Imaging		E1	Method
Goncalves et al. [2020]	EHR		C2	Study
Ho et al. [2020]	General		E1	Method
Yoon et al. [2020]	EHR		C2	Method
Zhang et al. [2020]	EHR		C2	Method
Chaitanya et al. [2021]	Imaging	P4	E1	Method
Chen et al. [2021a]	Multimodal		C2, O1	Review
Chlap et al. [2021]	Imaging		E1	Review
Jordon et al. [2022]	General		C2	Review
Stadler et al. [2022]	Tabular		C2	Study
van den Goorbergh et al. [2022]	Tabular		O2	Study
Borisov et al. [2023]	Tabular	P2	C2	Method
Kazerouni et al. [2023]	Imaging		E1	Review
Khader et al. [2023]	Imaging		E1, C2	Method
Kotelnikov et al. [2023]	Tabular		C2	Method
Ktena et al. [2024]	Imaging		O3, T2	Method
Shumailov et al. [2024]	General		O3	Evidence
No remedy: documented scarcity, failure or risk (33)				
El Emam et al. [2011]	Tabular		C2	Review
Button et al. [2013]	General		O2	Evidence
Shokri et al. [2017]	General		C2	Evidence
Varoquaux [2018]	Imaging		O1, O2	Evidence
Zech et al. [2018]	Imaging		T2	Evidence
Bagdasaryan et al. [2019]	General		O3, C2	Evidence
Brasil et al. [2019]	Multimodal		O1	Review
Christodoulou et al. [2019]	Tabular		O2	Review
Jayaraman and Evans [2019]	General		C2	Evidence
Kim et al. [2019]	Imaging		T2	Review
Liu et al. [2019]	Imaging		T2	Review
Obermeyer et al. [2019]	EHR		O3	Evidence
Rocher et al. [2019]	Tabular		C2	Evidence
Schwarz et al. [2019]	Imaging		C2	Evidence
Zhu et al. [2019]	General		C1, C2	Evidence

Work	Modality	Sec.	Scarcity	Type
Geiping et al. [2020]	General		C1, C2	Evidence
Nagendran et al. [2020]	Imaging		T2	Review
Nguengang Wakap et al. [2020]	Tabular		O1	Evidence
Oakden-Rayner et al. [2020]	Imaging		O3	Evidence
Schaefer et al. [2020]	Multimodal		O1	Review
Willemink et al. [2020]	Imaging		E1, E2, C2	Review
Wynants et al. [2020]	Multimodal		O2, T1	Evidence
Chen et al. [2021b]	Multimodal		O3	Review
Finlayson et al. [2021]	Multimodal		T1, T2	Evidence
Koh et al. [2021]	Multimodal		T2	Resource
Roberts et al. [2021]	Imaging		O2, T1	Evidence
Wong et al. [2021]	EHR		T2	Evidence
Carlini et al. [2022]	General		C2	Evidence
Gichoya et al. [2022]	Imaging		O3, T2	Evidence
Packhäuser et al. [2022]	Imaging		C2	Evidence
Varoquaux and Cheplygina [2022]	Imaging		O1, T2	Evidence
Yu et al. [2022]	Imaging		T2	Review
Hatamizadeh et al. [2023]	Imaging		C1, C2	Evidence

REFERENCES

Martin Abadi, Andy Chu, Ian Goodfellow, H. Brendan McMahan, Ilya Mironov, Kunal Talwar, and Li Zhang. 2016. Deep Learning with Differential Privacy. In *Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security (CCS)*. 308-318. https://doi.org/10.1145/2976749.2978318

Mohammed Adnan, Shivam Kalra, Jesse C. Cresswell, Graham W. Taylor, and Hamid R. Tizhoosh. 2022. Federated learning and differential privacy for medical image analysis. *Scientific Reports* 12 (2022), 1953. https://doi.org/10.1038/s41598-022-05539-7

Vibhu Agarwal, Tanya Podchiyska, Juan M. Banda, Veena Goel, Tiffany I. Leung, Evan P. Minty, Timothy E. Sweeney, Elsie Gyang, and Nigam H. Shah. 2016. Learning statistical models of phenotypes using noisy labeled training data. *Journal of the American Medical Informatics Association* 23, 6 (2016), 1166-1173. https://doi.org/10.1093/jamia/ocw028

Monica Agrawal, Stefan Hegselmann, Hunter Lang, Yoon Kim, and David Sontag. 2022. Large language models are few-shot clinical information extractors. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics, 1998–2022. https://doi.org/10.18653/v1/2022.emnlp-main.130

Emily Alsentzer, John Murphy, William Boag, Wei-Hung Weng, Di Jindi, Tristan Naumann, and Matthew McDermott. 2019. Publicly Available Clinical BERT Embeddings. In *Proceedings of the 2nd Clinical Natural Language Processing Workshop*. 72-78.

Martin Arjovsky, Léon Bottou, Ishaan Gulrajani, and David Lopez-Paz. 2019. Invariant Risk Minimization. *arXiv preprint arXiv:1907.02893* (2019).

Shekoofeh Azizi, Basil Mustafa, Fiona Ryan, Zachary Beaver, Jan Freyberg, Jonathan Deaton, et al. 2021. Big Self-Supervised Models Advance Medical Image Classification. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. 3478-3488.

Eugene Bagdasaryan, Omid Poursaeed, and Vitaly Shmatikov. 2019. Differential Privacy Has Disparate Impact on Model Accuracy. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 32.

Wenjia Bai, Ozan Oktay, Matthew Sinclair, Hideaki Suzuki, Martin Rajchl, Giacomo Tarroni, Ben Glocker, Andrew King, Paul M. Matthews, and Daniel Rueckert. 2017. Semi-supervised Learning for Network-Based Cardiac MR Image Segmentation. In *Medical Image Computing and Computer-Assisted Intervention (MICCAI) (LNCS, Vol. 10434)*. 253-260.

Xiang Bai, Hanchen Wang, Liya Ma, Yongchao Xu, Jiefeng Gan, Ziwei Fan, et al. 2021. Advancing COVID-19 diagnosis with privacy-preserving collaboration in artificial intelligence. *Nature Machine Intelligence* 3 (2021), 1081-1089. https://doi.org/10.1038/s42256-021-00421-z

Indranil Balki, Afsaneh Amirabadi, Jacob Levman, Anne L. Martel, Ziga Emersic, Blaz Meden, et al. 2019. Sample-Size Determination Methodologies for Machine Learning in Medical Imaging Research: A Systematic Review. *Canadian Association of Radiologists Journal* 70, 4 (2019), 344-353. https://doi.org/10.1016/j.carj.2019.06.002

Jineta Banerjee, Jaclyn N. Taroni, Robert J. Allaway, Deepashree Venkatesh Prasad, Justin Guinney, and Casey Greene. 2023. Machine learning in rare disease. *Nature Methods* 20 (2023), 803-814.

Mrinal Kanti Baowaly, Chia-Ching Lin, Chao-Lin Liu, and Kuan-Ta Chen. 2019. Synthesizing electronic health records using improved generative adversarial networks. *Journal of the American Medical Informatics Association* 26, 3 (2019), 228-241. https://doi.org/10.1093/jamia/ocy142

Jonathan Baxter. 2000. A Model of Inductive Bias Learning. *Journal of Artificial Intelligence Research* 12 (2000), 149-198. https://doi.org/10.1613/jair.731

Brett K. Beaulieu-Jones, Zhiwei Steven Wu, Chris Williams, Ran Lee, Sanjeev P. Bhavnani, James Brian Byrd, and Casey S. Greene. 2019. Privacy-Preserving Generative Deep Neural Networks Support Clinical Data Sharing. *Circulation: Cardiovascular Quality and Outcomes* 12, 7 (2019), e005122. https://doi.org/10.1161/CIRCOUTCOMES.118.005122

Aurélien Bellet, Rachid Guerraoui, Mahsa Taziki, and Marc Tommasi. 2018. Personalized and Private Peer-to-Peer Machine Learning. In *Proceedings of the 21st International Conference on Artificial Intelligence and Statistics (AISTATS) (PMLR, Vol. 84)*.

Shai Ben-David, John Blitzer, Koby Crammer, Alex Kulesza, Fernando Pereira, and Jennifer Wortman Vaughan. 2010. A theory of learning from different domains. *Machine Learning* 79 (2010), 151-175. https://doi.org/10.1007/s10994-009-5152-4

Oya Beyan, Ananya Choudhury, Johan van Soest, Oliver Kohlbacher, Lukas Zimmermann, Holger Stenzhorn, et al. 2020. Distributed Analytics on Sensitive Medical Data: The Personal Health Train. *Data Intelligence* 2, 1-2 (2020), 96-107. https://doi.org/10.1162/dint_a_00032

Peva Blanchard, El Mahdi El Mhamdi, Rachid Guerraoui, and Julien Stainer. 2017. Machine Learning with Adversaries: Byzantine Tolerant Gradient Descent. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 30.

Keith Bonawitz, Vladimir Ivanov, Ben Kreuter, Antonio Marcedone, H. Brendan McMahan, Sarvar Patel, Daniel Ramage, Aaron Segal, and Karn Seth. 2017. Practical Secure Aggregation for Privacy-Preserving Machine Learning. In *Proceedings of the 2017 ACM SIGSAC Conference on Computer and Communications Security (CCS)*. 1175-1191. https://doi.org/10.1145/3133956.3133982

Vadim Borisov, Kathrin Seßler, Tobias Leemann, Martin Pawelczyk, and Gjergji Kasneci. 2023. Language Models are Realistic Tabular Data Generators. In *International Conference on Learning Representations (ICLR)*.

Christopher Bowles, Liang Chen, Ricardo Guerrero, Paul Bentley, Roger Gunn, Alexander Hammers, David Alexander Dickie, Maria Valdés Hernández, Joanna Wardlaw, and Daniel Rueckert. 2018. GAN Augmentation: Augmenting Training Data using Generative Adversarial Networks. *arXiv preprint arXiv:1810.10863* (2018).

Stephen Boyd, Arpita Ghosh, Balaji Prabhakar, and Devavrat Shah. 2006. Randomized Gossip Algorithms. *IEEE Transactions on Information Theory* 52, 6 (2006), 2508-2530. https://doi.org/10.1109/TIT.2006.874516

Sandra Brasil, Carlota Pascoal, Rita Francisco, Vanessa dos Reis Ferreira, Paula A. Videira, and Gonçalo Valadão. 2019. Artificial Intelligence (AI) in Rare Diseases: Is the Future Brighter? *Genes* 10, 12 (2019), 978. https://doi.org/10.3390/genes10120978

Theodora S. Brisimi, Ruidi Chen, Theofanie Mela, Alex Olshevsky, Ioannis Ch. Paschalidis, and Wei Shi. 2018. Federated learning of predictive models from federated Electronic Health Records. *International Journal of Medical Informatics* 112 (2018), 59-67. https://doi.org/10.1016/j.ijmedinf.2018.01.007

Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, et al. 2020. Language Models are Few-Shot Learners. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 33.

Samuel Budd, Emma C. Robinson, and Bernhard Kainz. 2021. A survey on active learning and human-in-the-loop deep learning for medical image analysis. *Medical Image Analysis* 71 (2021), 102062. https://doi.org/10.1016/j.media.2021.102062

Katherine S. Button, John P. A. Ioannidis, Claire Mokrysz, Brian A. Nosek, Jonathan Flint, Emma S. J. Robinson, and Marcus R. Munafò. 2013. Power failure: why small sample size undermines the reliability of neuroscience. *Nature Reviews Neuroscience* 14 (2013), 365-376. https://doi.org/10.1038/nrn3475

Gabriele Campanella, Matthew G. Hanna, Luke Geneslaw, Allen Miraflor, Vitor Werneck Krauss Silva, Klaus J. Busam, Edi Brogi, Victor E. Reuter, David S. Klimstra, and Thomas J. Fuchs. 2019. Clinical-grade computational pathology using weakly supervised deep learning on whole slide images. *Nature Medicine* 25 (2019), 1301-1309. https://doi.org/10.1038/s41591-019-0508-1

Nicholas Carlini, Steve Chien, Milad Nasr, Shuang Song, Andreas Terzis, and Florian Tramèr. 2022. Membership Inference Attacks From First Principles. In *IEEE Symposium on Security and Privacy (S&P)*. 1897-1914. https://doi.org/10.1109/SP46214.2022.9833649

Daniel C. Castro, Ian Walker, and Ben Glocker. 2020. Causality matters in medical imaging. *Nature Communications* 11 (2020), 3673. https://doi.org/10.1038/s41467-020-17478-w

Krishna Chaitanya, Ertunc Erdil, Neerav Karani, and Ender Konukoglu. 2020. Contrastive learning of global and local features for medical image segmentation with limited annotations. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 33.

Krishna Chaitanya, Neerav Karani, Christian F. Baumgartner, Ertunc Erdil, Anton Becker, Olivio Donati, and Ender Konukoglu. 2021. Semi-supervised task-driven data augmentation for medical image segmentation. *Medical Image Analysis* 68 (2021), 101934. https://doi.org/10.1016/j.media.2020.101934

Ken Chang, Niranjan Balachandar, Carson Lam, Darvin Yi, James Brown, Andrew Beers, Bruce Rosen, Daniel L. Rubin, and Jayashree Kalpathy-Cramer. 2018. Distributed deep learning networks among institutions for medical imaging. *Journal of the American Medical Informatics Association* 25, 8 (2018), 945-954. https://doi.org/10.1093/jamia/ocy017

Nitesh V. Chawla, Kevin W. Bowyer, Lawrence O. Hall, and W. Philip Kegelmeyer. 2002. SMOTE: Synthetic Minority Over-sampling Technique. *Journal of Artificial Intelligence Research* 16 (2002), 321-357. https://doi.org/10.1613/jair.953

Irene Y. Chen, Emma Pierson, Sherri Rose, Shalmali Joshi, Kadija Ferryman, and Marzyeh Ghassemi. 2021b. Ethical Machine Learning in Healthcare. *Annual Review of Biomedical Data Science* 4 (2021), 123-144. https://doi.org/10.1146/annurev-biodatasci-092820-114757

Richard J. Chen, Tong Ding, Ming Y. Lu, Drew F. K. Williamson, Guillaume Jaume, Andrew H. Song, et al. 2024. Towards a general-purpose foundation model for computational pathology. *Nature Medicine* 30 (2024), 850-862. https://doi.org/10.1038/s41591-024-02857-3

Richard J. Chen, Ming Y. Lu, Tiffany Y. Chen, Drew F. K. Williamson, and Faisal Mahmood. 2021a. Synthetic data in machine learning for medicine and healthcare. *Nature Biomedical Engineering* 5 (2021), 493-497. https://doi.org/10.1038/s41551-021-00751-8

Sihong Chen, Kai Ma, and Yefeng Zheng. 2019. Med3D: Transfer Learning for 3D Medical Image Analysis. *arXiv preprint arXiv:1904.00625* (2019).

Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. 2020a. A Simple Framework for Contrastive Learning of Visual Representations. In *Proceedings of the 37th International Conference on Machine Learning (ICML) (PMLR, Vol. 119)*. 1597-1607.

Yiqiang Chen, Xin Qin, Jindong Wang, Chaohui Yu, and Wen Gao. 2020b. FedHealth: A Federated Transfer Learning Framework for Wearable Healthcare. *IEEE Intelligent Systems* 35, 4 (2020), 83-93. https://doi.org/10.1109/MIS.2020.2988604

Veronika Cheplygina, Marleen de Bruijne, and Josien P. W. Pluim. 2019. Not-so-supervised: A survey of semi-supervised, multi-instance, and transfer learning in medical image analysis. *Medical Image Analysis* 54 (2019), 280-296. https://doi.org/10.1016/j.media.2019.03.009

Phillip Chlap, Hang Min, Nym Vandenberg, Jason Dowling, Lois Holloway, and Annette Haworth. 2021. A review of medical image data augmentation techniques for deep learning applications. *Journal of Medical Imaging and Radiation Oncology* 65, 5 (2021), 545-563. https://doi.org/10.1111/1754-9485.13261

Hyunghoon Cho, David J. Wu, and Bonnie Berger. 2018. Secure genome-wide association analysis using multiparty computation. *Nature Biotechnology* 36 (2018), 547-551. https://doi.org/10.1038/nbt.4108

Junghwan Cho, Kyewook Lee, Ellie Shin, Garry Choy, and Synho Do. 2015. How much data is needed to train a medical image deep learning system to achieve necessary high accuracy? *arXiv preprint arXiv:1511.06348* (2015).

Edward Choi, Mohammad Taha Bahadori, Andy Schuetz, Walter F. Stewart, and Jimeng Sun. 2016. Doctor AI: Predicting Clinical Events via Recurrent Neural Networks. In *Proceedings of the 1st Machine Learning for Healthcare Conference (MLHC) (PMLR, Vol. 56)*. 301-318.

Edward Choi, Mohammad Taha Bahadori, Le Song, Walter F. Stewart, and Jimeng Sun. 2017a. GRAM: Graph-based Attention Model for Healthcare Representation Learning. In *Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*. 787-795. https://doi.org/10.1145/3097983.3098126

Edward Choi, Siddharth Biswal, Bradley Malin, Jon Duke, Walter F. Stewart, and Jimeng Sun. 2017b. Generating Multi-label Discrete Patient Records using Generative Adversarial Networks. In *Proceedings of the 2nd Machine Learning for Healthcare Conference (MLHC) (PMLR, Vol. 68)*. 286-305.

Olivia Choudhury, Aris Gkoulalas-Divanis, Theodoros Salonidis, Issa Sylla, Yoonyoung Park, Grace Hsu, and Amar Das. 2019. Differential Privacy-enabled Federated Learning for Sensitive Health Data. *arXiv preprint arXiv:1910.02578* (2019).

Evangelia Christodoulou, Jie Ma, Gary S. Collins, Ewout W. Steyerberg, Jan Y. Verbakel, and Ben Van Calster. 2019. A systematic review shows no performance benefit of machine learning over logistic regression for clinical prediction models. *Journal of Clinical Epidemiology* 110 (2019), 12-22. https://doi.org/10.1016/j.jclinepi.2019.02.004

Liam Collins, Hamed Hassani, Aryan Mokhtari, and Sanjay Shakkottai. 2021. Exploiting Shared Representations for Personalized Federated Learning. In *Proceedings of the 38th International Conference on Machine Learning (ICML) (PMLR, Vol. 139)*. 2089-2099.

Haotian Cui, Chloe Wang, Hassaan Maan, Kuan Pang, Fengning Luo, Nan Duan, and Bo Wang. 2024. scGPT: toward building a foundation model for single-cell multi-omics using generative AI. *Nature Methods* 21 (2024), 1470-1480. https://doi.org/10.1038/s41592-024-02201-0

Edwige Cyffers and Aurélien Bellet. 2022. Privacy Amplification by Decentralization. In *Proceedings of the 25th International Conference on Artificial Intelligence and Statistics (AISTATS) (PMLR, Vol. 151)*.

Ittai Dayan, Holger R. Roth, Aoxiao Zhong, Ahmed Harouni, Amilcare Gentili, Anas Z. Abidin, et al. 2021. Federated learning for predicting clinical outcomes in patients with COVID-19. *Nature Medicine* 27 (2021), 1735-1743. https://doi.org/10.1038/s41591-021-01506-3

Timo M. Deist, Frank J. W. M. Dankers, Priyanka Ojha, et al. 2020. Distributed learning on 20 000+ lung cancer patients - The Personal Health Train. *Radiotherapy and Oncology* 144 (2020), 189-200. https://doi.org/10.1016/j.radonc.2019.11.019

Rebecca DerSimonian and Nan Laird. 1986. Meta-analysis in clinical trials. *Controlled Clinical Trials* 7, 3 (1986), 177-188. https://doi.org/10.1016/0197-2456(86)90046-2

Qi Dou, Tiffany Y. So, Meirui Jiang, Quande Liu, Varut Vardhanabhuti, Georgios Kaissis, et al. 2021. Federated deep learning for detecting COVID-19 lung abnormalities in CT: a privacy-preserving multinational validation study. *npj Digital*

Medicine 4 (2021), 60. https://doi.org/10.1038/s41746-021-00431-6

Simon S. Du, Wei Hu, Sham M. Kakade, Jason D. Lee, and Qi Lei. 2021. Few-Shot Learning via Learning the Representation, Provably. In *International Conference on Learning Representations (ICLR)*.

Rui Duan, Mary Regina Boland, Zixuan Liu, et al. 2020a. Learning from electronic health records across multiple sites: A communication-efficient and privacy-preserving distributed algorithm. *Journal of the American Medical Informatics Association* 27, 3 (2020), 376-385. https://doi.org/10.1093/jamia/ocz199

Rui Duan, Chongliang Luo, Martijn J. Schuemie, et al. 2020b. Learning from local to global: An efficient distributed algorithm for modeling time-to-event data. *Journal of the American Medical Informatics Association* 27, 7 (2020), 1028-1036. https://doi.org/10.1093/jamia/ocaa044

Cynthia Dwork, Frank McSherry, Kobbi Nissim, and Adam Smith. 2006. Calibrating Noise to Sensitivity in Private Data Analysis. In *Theory of Cryptography Conference (TCC) (LNCS, Vol. 3876)*. 265-284. https://doi.org/10.1007/11681878_14

Bradley Efron and Carl Morris. 1973. Stein's Estimation Rule and Its Competitors: An Empirical Bayes Approach. *J. Amer. Statist. Assoc.* 68, 341 (1973), 117-130. https://doi.org/10.1080/01621459.1973.10481350

Khaled El Emam, Elizabeth Jonker, Luk Arbuckle, and Bradley Malin. 2011. A Systematic Review of Re-Identification Attacks on Health Data. *PLoS ONE* 6, 12 (2011), e28071. https://doi.org/10.1371/journal.pone.0028071

Andre Esteva, Brett Kuprel, Roberto A. Novoa, Justin Ko, Susan M. Swetter, Helen M. Blau, and Sebastian Thrun. 2017. Dermatologist-level classification of skin cancer with deep neural networks. *Nature* 542 (2017), 115-118. https://doi.org/10.1038/nature21056

Alireza Fallah, Aryan Mokhtari, and Asuman Ozdaglar. 2020. Personalized Federated Learning with Theoretical Guarantees: A Model-Agnostic Meta-Learning Approach. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 33.

Alberto Fernández, Salvador García, Francisco Herrera, and Nitesh V. Chawla. 2018. SMOTE for Learning from Imbalanced Data: Progress and Challenges, Marking the 15-year Anniversary. *Journal of Artificial Intelligence Research* 61 (2018), 863-905. https://doi.org/10.1613/jair.1.11192

Rosa L. Figueroa, Qing Zeng-Treitler, Sasikiran Kandula, and Long H. Ngo. 2012. Predicting sample size required for classification performance. *BMC Medical Informatics and Decision Making* 12 (2012), 8. https://doi.org/10.1186/1472-6947-12-8

Samuel G. Finlayson, Adarsh Subbaswamy, Karandeep Singh, John Bowers, Annabel Kupke, Jonathan Zittrain, Isaac S. Kohane, and Suchi Saria. 2021. The Clinician and Dataset Shift in Artificial Intelligence. *New England Journal of Medicine* 385, 3 (2021), 283-286. https://doi.org/10.1056/NEJMc2104626

Chelsea Finn, Pieter Abbeel, and Sergey Levine. 2017. Model-Agnostic Meta-Learning for Fast Adaptation of Deep Networks. In *Proceedings of the 34th International Conference on Machine Learning (ICML) (PMLR, Vol. 70)*. 1126-1135.

Maayan Frid-Adar, Idit Diamant, Eyal Klang, Michal Amitai, Jacob Goldberger, and Hayit Greenspan. 2018. GAN-based synthetic medical image augmentation for increased CNN performance in liver lesion classification. *Neurocomputing* 321 (2018), 321-331. https://doi.org/10.1016/j.neucom.2018.09.013

Jason A. Fries, Paroma Varma, Vincent S. Chen, Ke Xiao, Heliodoro Tejeda, Priyanka Saha, et al. 2019. Weakly supervised classification of aortic valve malformations using unlabeled cardiac MRI sequences. *Nature Communications* 10 (2019), 3111. https://doi.org/10.1038/s41467-019-11012-3

David Froelicher, Juan R. Troncoso-Pastoriza, Jean Louis Raisaro, Michel A. Cuendet, Joao Sa Sousa, Hyunghoon Cho, Bonnie Berger, Jacques Fellay, and Jean-Pierre Hubaux. 2021. Truly privacy-preserving federated analytics for precision medicine with multiparty homomorphic encryption. *Nature Communications* 12 (2021), 5910. https://doi.org/10.1038/s41467-021-25972-y

Yarin Gal, Riashat Islam, and Zoubin Ghahramani. 2017. Deep Bayesian Active Learning with Image Data. In *Proceedings of the 34th International Conference on Machine Learning (ICML) (PMLR, Vol. 70)*. 1183-1192.

Amadou Gaye, Yannick Marcon, Julia Isaeva, Philippe LaFlamme, Andrew Turner, Elinor M. Jones, et al. 2014. DataSHIELD: taking the analysis to the data, not the data to the analysis. *International Journal of Epidemiology* 43, 6 (2014), 1929-1944. https://doi.org/10.1093/ije/dyu188

Jonas Geiping, Hartmut Bauermeister, Hannah Dröge, and Michael Moeller. 2020. Inverting Gradients - How easy is it to break privacy in federated learning?. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 33.

Andrew Gelman, Aleks Jakulin, Maria Grazia Pittau, and Yu-Sung Su. 2008. A weakly informative default prior distribution for logistic and other regression models. *The Annals of Applied Statistics* 2, 4 (2008), 1360–1383. https://doi.org/10.1214/08-AOAS191

Craig Gentry. 2009. Fully Homomorphic Encryption Using Ideal Lattices. In *Proceedings of the 41st Annual ACM Symposium on Theory of Computing (STOC)*. 169-178. https://doi.org/10.1145/1536414.1536440

Amirata Ghorbani and James Zou. 2019. Data Shapley: Equitable Valuation of Data for Machine Learning. In *Proceedings of the 36th International Conference on Machine Learning (ICML) (PMLR, Vol. 97)*. 2242-2251.

Avishek Ghosh, Jichan Chung, Dong Yin, and Kannan Ramchandran. 2020. An Efficient Framework for Clustered Federated Learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 33.

Judy Wawira Gichoya, Imon Banerjee, Ananth Reddy Bhimireddy, John L. Burns, Leo Anthony Celi, et al. 2022. AI recognition of patient race in medical imaging: a modelling study. *The Lancet Digital Health* 4, 6 (2022), e406-e414. https://doi.org/10.1016/S2589-7500(22)00063-2

Andre Goncalves, Priyadip Ray, Braden Soper, Jennifer Stevens, Linda Coyle, and Ana Paula Sales. 2020. Generation and evaluation of synthetic patient data. *BMC Medical Research Methodology* 20 (2020), 108. https://doi.org/10.1186/s12874-020-00977-1

Ian Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. 2014. Generative Adversarial Nets. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 27.

Hao Guan and Mingxia Liu. 2022. Domain Adaptation for Medical Image Analysis: A Survey. *IEEE Transactions on Biomedical Engineering* 69, 3 (2022), 1173-1185. https://doi.org/10.1109/TBME.2021.3117407

Ishaan Gulrajani and David Lopez-Paz. 2021. In Search of Lost Domain Generalization. In *International Conference on Learning Representations (ICLR)*.

Varun Gulshan, Lily Peng, Marc Coram, Martin C. Stumpe, Derek Wu, Arunachalam Narayanaswamy, et al. 2016. Development and Validation of a Deep Learning Algorithm for Detection of Diabetic Retinopathy in Retinal Fundus Photographs. *JAMA* 316, 22 (2016), 2402-2410. https://doi.org/10.1001/jama.2016.17216

Pengfei Guo, Puyang Wang, Jinyuan Zhou, Shanshan Jiang, and Vishal M. Patel. 2021. Multi-institutional Collaborations for Improving Deep Learning-based Magnetic Resonance Image Reconstruction Using Federated Learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2423-2432.

Otkrist Gupta and Ramesh Raskar. 2018. Distributed learning of deep neural network over multiple agents. *Journal of Network and Computer Applications* 116 (2018), 1-8. https://doi.org/10.1016/j.jnca.2018.05.003

Yaron Gurovich, Yair Hanani, Omri Bar, Guy Nadav, Nicole Fleischer, Dekel Gelbman, et al. 2019. Identifying facial phenotypes of genetic disorders using deep learning. *Nature Medicine* 25 (2019), 60-64. https://doi.org/10.1038/s41591-018-0279-0

Yoni Halpern, Steven Horng, Youngduck Choi, and David Sontag. 2016. Electronic medical record phenotyping using the anchor and learn framework. *Journal of the American Medical Informatics Association* 23, 4 (2016), 731-740. https://doi.org/10.1093/jamia/ocw011

Ali Hatamizadeh, Hongxu Yin, Pavlo Molchanov, Andriy Myronenko, Wenqi Li, Prerna Dogra, et al. 2023. Do Gradient Inversion Attacks Make Federated Learning Unsafe? *IEEE Transactions on Medical Imaging* 42, 7 (2023), 2044-2056. https://doi.org/10.1109/TMI.2023.3239391

Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. 2022. Masked Autoencoders Are Scalable Vision Learners. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 16000-16009.

Stefan Hegselmann, Alejandro Buendia, Hunter Lang, Monica Agrawal, Xiaoyi Jiang, and David Sontag. 2023. TabLLM: Few-shot Classification of Tabular Data with Large Language Models. In *Proceedings of the 26th International Conference on Artificial Intelligence and Statistics (AISTATS) (PMLR, Vol. 206)*. 5549-5581.

Jonathan Ho, Ajay Jain, and Pieter Abbeel. 2020. Denoising Diffusion Probabilistic Models. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 33.

Noah Hollmann, Samuel Müller, Katharina Eggensperger, and Frank Hutter. 2023. TabPFN: A Transformer That Solves Small Tabular Classification Problems in a Second. In *International Conference on Learning Representations (ICLR)*.

Noah Hollmann, Samuel Müller, Lennart Purucker, Arjun Krishnakumar, Max Körfer, Shi Bin Hoo, Robin Tibor Schirrmeister, and Frank Hutter. 2025. Accurate predictions on small data with a tabular foundation model. *Nature* 637 (2025), 319-326. https://doi.org/10.1038/s41586-024-08328-6

Timothy Hospedales, Antreas Antoniou, Paul Micaelli, and Amos Storkey. 2022. Meta-Learning in Neural Networks: A Survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 44, 9 (2022), 5149-5169. https://doi.org/10.1109/TPAMI.2021.3079209

George Hripcsak, Jon D. Duke, Nigam H. Shah, Christian G. Reich, Vojtech Huser, Martijn J. Schuemie, et al. 2015. Observational Health Data Sciences and Informatics (OHDSI): Opportunities for Observational Researchers. *Studies in Health Technology and Informatics* 216 (2015), 574-578.

Kevin Hsieh, Amar Phanishayee, Onur Mutlu, and Phillip Gibbons. 2020. The Non-IID Data Quagmire of Decentralized Machine Learning. In *Proceedings of the 37th International Conference on Machine Learning (ICML) (PMLR, Vol. 119)*. 4387-4398.

Tzung-Chien Hsieh, Aviram Bar-Haim, Shahida Moosa, Nadja Ehmke, Karen W. Gripp, Jean Tori Pantel, et al. 2022. GestaltMatcher facilitates rare disease matching using facial phenotype descriptors. *Nature Genetics* 54 (2022), 349-357. https://doi.org/10.1038/s41588-021-01010-x

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. LoRA: Low-Rank Adaptation of Large Language Models. In *International Conference on Learning Representations (ICLR)*.

Li Huang, Andrew L. Shea, Huining Qian, Aditya Masurkar, Hao Deng, and Dianbo Liu. 2019. Patient clustering improves efficiency of federated machine learning to predict mortality and hospital stay time using distributed electronic medical

records. *Journal of Biomedical Informatics* 99 (2019), 103291. https://doi.org/10.1016/j.jbi.2019.103291

Shih-Cheng Huang, Liyue Shen, Matthew P. Lungren, and Serena Yeung. 2021b. GLoRIA: A Multimodal Global-Local Representation Learning Framework for Label-efficient Medical Image Recognition. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. 3942-3951.

Yutao Huang, Lingyang Chu, Zirui Zhou, Lanjun Wang, Jiangchuan Liu, Jian Pei, and Yong Zhang. 2021a. Personalized Cross-Silo Federated Learning on Non-IID Data. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 35. 7865-7873.

Maximilian Ilse, Jakub Tomczak, and Max Welling. 2018. Attention-based Deep Multiple Instance Learning. In *Proceedings of the 35th International Conference on Machine Learning (ICML) (PMLR, Vol. 80)*. 2127-2136.

Jeremy Irvin, Pranav Rajpurkar, Michael Ko, Yifan Yu, Silviana Ciurea-Ilcus, Chris Chute, et al. 2019. CheXpert: A Large Chest Radiograph Dataset with Uncertainty Labels and Expert Comparison. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 33. 590-597.

Bargav Jayaraman and David Evans. 2019. Evaluating Differentially Private Machine Learning in Practice. In *28th USENIX Security Symposium*. 1895-1912.

Lavender Yao Jiang, Xujin Chris Liu, Nima Pour Nejatian, Mustafa Nasir-Moin, Duo Wang, Anas Abidin, et al. 2023. Health system-scale language models are all-purpose prediction engines. *Nature* 619 (2023), 357-362. https://doi.org/10.1038/s41586-023-06160-y

Meirui Jiang, Zirui Wang, and Qi Dou. 2022. HarmoFL: Harmonizing Local and Global Drifts in Federated Learning on Heterogeneous Medical Images. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 36. 1087-1095.

Wenchao Jiang, Pinghao Li, Shuang Wang, Yuan Wu, Meng Xue, Lucila Ohno-Machado, and Xiaoqian Jiang. 2013. WebGLORE: a web service for Grid LOgistic REgression. *Bioinformatics* 29, 24 (2013), 3238-3240. https://doi.org/10.1093/bioinformatics/btt559

Arthur Jochems, Timo M. Deist, Issam El Naqa, et al. 2017. Developing and Validating a Survival Prediction Model for NSCLC Patients Through Distributed Learning Across 3 Countries. *International Journal of Radiation Oncology, Biology, Physics* 99, 2 (2017), 344-352. https://doi.org/10.1016/j.ijrobp.2017.04.021

Arthur Jochems, Timo M. Deist, Johan van Soest, et al. 2016. Distributed learning: Developing a predictive model based on data from multiple hospitals without data leaving the hospital - A real life proof of concept. *Radiotherapy and Oncology* 121, 3 (2016), 459-467. https://doi.org/10.1016/j.radonc.2016.10.002

Alistair E. W. Johnson, Tom J. Pollard, Lu Shen, Li-wei H. Lehman, Mengling Feng, Mohammad Ghassemi, Benjamin Moody, Peter Szolovits, Leo Anthony Celi, and Roger G. Mark. 2016. MIMIC-III, a freely accessible critical care database. *Scientific Data* 3 (2016), 160035. https://doi.org/10.1038/sdata.2016.35

Michael I. Jordan, Jason D. Lee, and Yun Yang. 2019. Communication-Efficient Distributed Statistical Inference. *J. Amer. Statist. Assoc.* 114, 526 (2019), 668-681. https://doi.org/10.1080/01621459.2018.1429274

James Jordon, Lukasz Szpruch, Florimond Houssiau, Mirko Bottarelli, Giovanni Cherubin, Carsten Maple, Samuel N. Cohen, and Adrian Weller. 2022. Synthetic Data - what, why and how? *arXiv preprint arXiv:2205.03257* (2022).

James Jordon, Jinsung Yoon, and Mihaela van der Schaar. 2019. PATE-GAN: Generating Synthetic Data with Differential Privacy Guarantees. In *International Conference on Learning Representations (ICLR)*.

Peter Kairouz, H. Brendan McMahan, Brendan Avent, Aurélien Bellet, Mehdi Bennis, et al. 2021. Advances and Open Problems in Federated Learning. *Foundations and Trends in Machine Learning* 14, 1-2 (2021), 1-210. https://doi.org/10.1561/2200000083

Georgios Kaissis, Alexander Ziller, Jonathan Passerat-Palmbach, Théo Ryffel, Dmitrii Usynin, Andrew Trask, et al. 2021. End-to-end privacy preserving deep learning on multi-institutional medical imaging. *Nature Machine Intelligence* 3 (2021), 473-484. https://doi.org/10.1038/s42256-021-00337-8

Georgios A. Kaissis, Marcus R. Makowski, Daniel Rückert, and Rickmer F. Braren. 2020. Secure, privacy-preserving and federated machine learning in medical imaging. *Nature Machine Intelligence* 2 (2020), 305-311. https://doi.org/10.1038/s42256-020-0186-1

Shivam Kalra, Junfeng Wen, Jesse C. Cresswell, Maksims Volkovs, and H. R. Tizhoosh. 2023. Decentralized federated learning through proxy model sharing. *Nature Communications* 14, 1, Article 2899 (2023). https://doi.org/10.1038/s41467-023-38569-4

Konstantinos Kamnitsas, Christian Baumgartner, Christian Ledig, Virginia Newcombe, Joanna Simpson, Andrew Kane, et al. 2017. Unsupervised Domain Adaptation in Brain Lesion Segmentation with Adversarial Networks. In *Information Processing in Medical Imaging (IPMI) (LNCS, Vol. 10265)*. 597-609.

Sai Praneeth Karimireddy, Satyen Kale, Mehryar Mohri, Sashank Reddi, Sebastian Stich, and Ananda Theertha Suresh. 2020. SCAFFOLD: Stochastic Controlled Averaging for Federated Learning. In *Proceedings of the 37th International Conference on Machine Learning (ICML) (PMLR, Vol. 119)*. 5132-5143.

George Em Karniadakis, Ioannis G. Kevrekidis, Lu Lu, Paris Perdikaris, Sifan Wang, and Liu Yang. 2021. Physics-informed machine learning. *Nature Reviews Physics* 3 (2021), 422-440. https://doi.org/10.1038/s42254-021-00314-5

Amirhossein Kazerouni, Ehsan Khodapanah Aghdam, Moein Heidari, Reza Azad, Mohsen Fayyaz, Ilker Hacihaliloglu, and Dorit Merhof. 2023. Diffusion models in medical imaging: A comprehensive survey. *Medical Image Analysis* 88 (2023), 102846. https://doi.org/10.1016/j.media.2023.102846

Firas Khader, Gustav Müller-Franzes, Soroosh Tayebi Arasteh, Tianyu Han, Christoph Haarburger, Maximilian Schulze-Hagen, et al. 2023. Denoising diffusion probabilistic models for 3D medical image generation. *Scientific Reports* 13 (2023), 7303. https://doi.org/10.1038/s41598-023-34341-2

Dong Wook Kim, Hye Young Jang, Kyung Won Kim, Youngbin Shin, and Seong Ho Park. 2019. Design Characteristics of Studies Reporting the Performance of Artificial Intelligence Algorithms for Diagnostic Analysis of Medical Images: Results from Recently Published Papers. *Korean Journal of Radiology* 20, 3 (2019), 405-410. https://doi.org/10.3348/kjr.2019.0025

Yejin Kim, Jimeng Sun, Hwanjo Yu, and Xiaoqian Jiang. 2017. Federated Tensor Factorization for Computational Phenotyping. In *Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*. 887-895. https://doi.org/10.1145/3097983.3098118

Georgios Kissas, Yibo Yang, Eileen Hwuang, Walter R. Witschey, John A. Detre, and Paris Perdikaris. 2020. Machine learning in cardiovascular flows modeling: Predicting arterial blood pressure from non-invasive 4D flow MRI data using physics-informed neural networks. *Computer Methods in Applied Mechanics and Engineering* 358 (2020), 112623. https://doi.org/10.1016/j.cma.2019.112623

Pang Wei Koh, Shiori Sagawa, Henrik Marklund, Sang Michael Xie, Marvin Zhang, Akshay Balsubramani, et al. 2021. WILDS: A Benchmark of in-the-Wild Distribution Shifts. In *Proceedings of the 38th International Conference on Machine Learning (ICML) (PMLR, Vol. 139)*. 5637-5664.

Sebastian Köhler, Michael Gargano, Nicolas Matentzoglu, Leigh C. Carmody, David Lewis-Smith, Nicole A. Vasilevsky, et al. 2021. The Human Phenotype Ontology in 2021. *Nucleic Acids Research* 49, D1 (2021), D1207-D1217. https://doi.org/10.1093/nar/gkaa1043

Anastasia Koloskova, Nicolas Loizou, Sadra Boreiri, Martin Jaggi, and Sebastian U. Stich. 2020. A Unified Theory of Decentralized SGD with Changing Topology and Local Updates. In *Proceedings of the 37th International Conference on Machine Learning (ICML) (PMLR, Vol. 119)*. 5381-5393.

Akim Kotelnikov, Dmitry Baranchuk, Ivan Rubachev, and Artem Babenko. 2023. TabDDPM: Modelling Tabular Data with Diffusion Models. In *Proceedings of the 40th International Conference on Machine Learning (ICML) (PMLR, Vol. 202)*. 17564-17579.

Rayan Krishnan, Pranav Rajpurkar, and Eric J. Topol. 2022. Self-supervised learning in medicine and healthcare. *Nature Biomedical Engineering* 6 (2022), 1346-1352. https://doi.org/10.1038/s41551-022-00914-1

Ira Ktena, Olivia Wiles, Isabela Albuquerque, Sylvestre-Alvise Rebuffi, Ryutaro Tanno, Abhijit Guha Roy, et al. 2024. Generative models improve fairness of medical classifiers under distribution shifts. *Nature Medicine* 30 (2024), 1166-1173. https://doi.org/10.1038/s41591-024-02838-6

John H. Lagergren, John T. Nardini, Ruth E. Baker, Matthew J. Simpson, and Kevin B. Flores. 2020. Biologically-informed neural networks guide mechanistic modeling from sparse experimental data. *PLoS Computational Biology* 16, 12 (2020), e1008462. https://doi.org/10.1371/journal.pcbi.1008462

Brenden M. Lake, Ruslan Salakhutdinov, and Joshua B. Tenenbaum. 2015. Human-level concept learning through probabilistic program induction. *Science* 350, 6266 (2015), 1332-1338. https://doi.org/10.1126/science.aab3050

Junghye Lee, Jimeng Sun, Fei Wang, Shuang Wang, Chi-Hyuck Jun, and Xiaoqian Jiang. 2018. Privacy-Preserving Patient Similarity Learning in a Federated Environment: Development and Analysis. *JMIR Medical Informatics* 6, 2 (2018), e20. https://doi.org/10.2196/medinform.7744

Jinhyuk Lee, Wonjin Yoon, Sungdong Kim, Donghyeon Kim, Sunkyu Kim, Chan Ho So, and Jaewoo Kang. 2020. BioBERT: a pre-trained biomedical language representation model for biomedical text mining. *Bioinformatics* 36, 4 (2020), 1234-1240. https://doi.org/10.1093/bioinformatics/btz682

Tian Li, Shengyuan Hu, Ahmad Beirami, and Virginia Smith. 2021a. Ditto: Fair and Robust Federated Learning Through Personalization. In *Proceedings of the 38th International Conference on Machine Learning (ICML) (PMLR, Vol. 139)*. 6357-6368.

Tian Li, Anit Kumar Sahu, Manzil Zaheer, Maziar Sanjabi, Ameet Talwalkar, and Virginia Smith. 2020c. Federated Optimization in Heterogeneous Networks. In *Proceedings of Machine Learning and Systems (MLSys)*, Vol. 2. 429-450.

Wenqi Li, Fausto Milletarì, Daguang Xu, Nicola Rieke, Jonny Hancox, Wentao Zhu, et al. 2019. Privacy-Preserving Federated Brain Tumour Segmentation. In *Machine Learning in Medical Imaging (MLMI) (LNCS, Vol. 11861)*. 133-141.

Xiaoxiao Li, Yufeng Gu, Nicha Dvornek, Lawrence H. Staib, Pamela Ventola, and James S. Duncan. 2020a. Multi-site fMRI analysis using privacy-preserving federated learning and domain adaptation: ABIDE results. *Medical Image Analysis* 65 (2020), 101765. https://doi.org/10.1016/j.media.2020.101765

Xiaoxiao Li, Meirui Jiang, Xiaofei Zhang, Michael Kamp, and Qi Dou. 2021b. FedBN: Federated Learning on Non-IID Features via Local Batch Normalization. In *International Conference on Learning Representations (ICLR)*.

Xiaomeng Li, Lequan Yu, Hao Chen, Chi-Wing Fu, Lei Xing, and Pheng-Ann Heng. 2021c. Transformation-Consistent Self-Ensembling Model for Semisupervised Medical Image Segmentation. *IEEE Transactions on Neural Networks and Learning Systems* 32, 2 (2021), 523-534. https://doi.org/10.1109/TNNLS.2020.2995319

Yikuan Li, Shishir Rao, José Roberto Ayala Solares, Abdelaali Hassaine, Rema Ramakrishnan, Dexter Canoy, Yajie Zhu, Kazem Rahimi, and Gholamreza Salimi-Khorshidi. 2020b. BEHRT: Transformer for Electronic Health Records. *Scientific Reports* 10 (2020), 7155. https://doi.org/10.1038/s41598-020-62922-y

Xiangru Lian, Ce Zhang, Huan Zhang, Cho-Jui Hsieh, Wei Zhang, and Ji Liu. 2017. Can Decentralized Algorithms Outperform Centralized Algorithms? A Case Study for Decentralized Parallel Stochastic Gradient Descent. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 30.

Tao Lin, Lingjing Kong, Sebastian U. Stich, and Martin Jaggi. 2020. Ensemble Distillation for Robust Model Fusion in Federated Learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 33.

Quande Liu, Cheng Chen, Jing Qin, Qi Dou, and Pheng-Ann Heng. 2021. FedDG: Federated Domain Generalization on Medical Image Segmentation via Episodic Learning in Continuous Frequency Space. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 1013-1023.

Xiaoxuan Liu, Livia Faes, Aditya U. Kale, Siegfried K. Wagner, Dun Jack Fu, Alice Bruynseels, et al. 2019. A comparison of deep learning performance against health-care professionals in detecting diseases from medical imaging: a systematic review and meta-analysis. *The Lancet Digital Health* 1, 6 (2019), e271-e297. https://doi.org/10.1016/S2589-7500(19)30123-2

Charles Lu, Yaodong Yu, Sai Praneeth Karimireddy, Michael I. Jordan, and Ramesh Raskar. 2023. Federated Conformal Predictors for Distributed Uncertainty Quantification. In *Proceedings of the 40th International Conference on Machine Learning (ICML) (PMLR, Vol. 202)*.

Chia-Lun Lu, Shuang Wang, Zhanglong Ji, Yuan Wu, Li Xiong, Xiaoqian Jiang, and Lucila Ohno-Machado. 2015. WebDISCO: a web service for distributed cox model learning without patient-level data sharing. *Journal of the American Medical Informatics Association* 22, 6 (2015), 1212-1219. https://doi.org/10.1093/jamia/ocv083

James Lu, Kaiwen Deng, Xinyuan Zhang, Gengbo Liu, and Yuanfang Guan. 2021a. Neural-ODE for pharmacokinetics modeling and its advantage to alternative machine learning models in predicting new dosing regimens. *iScience* 24, 7 (2021), 102804. https://doi.org/10.1016/j.isci.2021.102804

Ming Y. Lu, Bowen Chen, Drew F. K. Williamson, Richard J. Chen, Ivy Liang, Tong Ding, et al. 2024. A visual-language foundation model for computational pathology. *Nature Medicine* 30 (2024), 863-874. https://doi.org/10.1038/s41591-024-02856-4

Ming Y. Lu, Richard J. Chen, Dehan Kong, Jana Lipkova, Rajendra Singh, Drew F. K. Williamson, Tiffany Y. Chen, and Faisal Mahmood. 2022. Federated learning for computational pathology on gigapixel whole slide images. *Medical Image Analysis* 76 (2022), 102298. https://doi.org/10.1016/j.media.2021.102298

Ming Y. Lu, Drew F. K. Williamson, Tiffany Y. Chen, Richard J. Chen, Matteo Barbieri, and Faisal Mahmood. 2021b. Data-efficient and weakly supervised computational pathology on whole-slide images. *Nature Biomedical Engineering* 5 (2021), 555-570. https://doi.org/10.1038/s41551-020-00682-w

Fenglong Ma, Quanzeng You, Houping Xiao, Radha Chitta, Jing Zhou, and Jing Gao. 2018. KAME: Knowledge-based Attention Model for Diagnosis Prediction in Healthcare. In *Proceedings of the 27th ACM International Conference on Information and Knowledge Management (CIKM)*. 743-752. https://doi.org/10.1145/3269206.3271701

Jun Ma, Yuting He, Feifei Li, Lin Han, Chenyu You, and Bo Wang. 2024. Segment anything in medical images. *Nature Communications* 15 (2024), 654. https://doi.org/10.1038/s41467-024-44824-z

Kushagra Mahajan, Monika Sharma, and Lovekesh Vig. 2020. Meta-DermDiagnosis: Few-Shot Skin Disease Identification Using Meta-Learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*. IEEE, 3142–3151. https://doi.org/10.1109/CVPRW50498.2020.00373

Joshua C. Mandel, David A. Kreda, Kenneth D. Mandl, Isaac S. Kohane, and Rachel B. Ramoni. 2016. SMART on FHIR: a standards-based, interoperable apps platform for electronic health records. *Journal of the American Medical Informatics Association* 23, 5 (2016), 899-908. https://doi.org/10.1093/jamia/ocv189

Othmane Marfoq, Giovanni Neglia, Aurélien Bellet, Laetitia Kameni, and Richard Vidal. 2021. Federated Multi-Task Learning under a Mixture of Distributions. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 34.

Andreas Maurer, Massimiliano Pontil, and Bernardino Romera-Paredes. 2016. The Benefit of Multitask Representation Learning. *Journal of Machine Learning Research* 17, 81 (2016), 1-32.

Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson, and Blaise Agüera y Arcas. 2017. Communication-Efficient Learning of Deep Networks from Decentralized Data. In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS) (PMLR, Vol. 54)*. 1273-1282.

H. Brendan McMahan, Daniel Ramage, Kunal Talwar, and Li Zhang. 2018. Learning Differentially Private Recurrent Language Models. In *International Conference on Learning Representations (ICLR)*.

Xueyan Mei, Zelong Liu, Philip M. Robson, Brett Marinelli, Mingqian Huang, Amish Doshi, et al. 2022. RadImageNet: An Open Radiologic Deep Learning Research Dataset for Effective Transfer Learning. *Radiology: Artificial Intelligence* 4, 5

(2022), e210315. https://doi.org/10.1148/ryai.210315

Ilya Mironov. 2017. Rényi Differential Privacy. In *IEEE 30th Computer Security Foundations Symposium (CSF)*. 263-275. https://doi.org/10.1109/CSF.2017.11

Mehryar Mohri, Gary Sivek, and Ananda Theertha Suresh. 2019. Agnostic Federated Learning. In *Proceedings of the 36th International Conference on Machine Learning (ICML) (PMLR, Vol. 97)*. 4615-4625.

Michael Moor, Oishi Banerjee, Zahra Shakeri Hossein Abad, Harlan M. Krumholz, Jure Leskovec, Eric J. Topol, and Pranav Rajpurkar. 2023. Foundation models for generalist medical artificial intelligence. *Nature* 616 (2023), 259-265. https://doi.org/10.1038/s41586-023-05881-4

Myura Nagendran, Yang Chen, Christopher A. Lovejoy, Anthony C. Gordon, Matthieu Komorowski, Hugh Harvey, Eric J. Topol, John P. A. Ioannidis, Gary S. Collins, and Mahiben Maruthappu. 2020. Artificial intelligence versus clinicians: systematic review of design, reporting standards, and claims of deep learning studies. *BMJ* 368 (2020), m689. https://doi.org/10.1136/bmj.m689

Vishwesh Nath, Dong Yang, Bennett A. Landman, Daguang Xu, and Holger R. Roth. 2021. Diminishing Uncertainty Within the Training Pool: Active Learning for Medical Image Segmentation. *IEEE Transactions on Medical Imaging* 40, 10 (2021), 2534-2547. https://doi.org/10.1109/TMI.2020.3048055

Angelia Nedić and Asuman Ozdaglar. 2009. Distributed Subgradient Methods for Multi-Agent Optimization. *IEEE Trans. Automat. Control* 54, 1 (2009), 48-61. https://doi.org/10.1109/TAC.2008.2009515

Stéphanie Nguengang Wakap, Deborah M. Lambert, Annie Olry, Charlotte Rodwell, Charlotte Gueydan, Valérie Lanneau, Daniel Murphy, Yann Le Cam, and Ana Rath. 2020. Estimating cumulative point prevalence of rare diseases: analysis of the Orphanet database. *European Journal of Human Genetics* 28 (2020), 165-173. https://doi.org/10.1038/s41431-019-0508-0

Dinh C. Nguyen, Quoc-Viet Pham, Pubudu N. Pathirana, Ming Ding, Aruna Seneviratne, Zihuai Lin, Octavia Dobre, and Won-Joo Hwang. 2022. Federated Learning for Smart Healthcare: A Survey. *Comput. Surveys* 55, 3 (2022), 60:1-60:37. https://doi.org/10.1145/3501296

Harsha Nori, Nicholas King, Scott Mayer McKinney, Dean Carignan, and Eric Horvitz. 2023. Capabilities of GPT-4 on Medical Challenge Problems. *arXiv preprint arXiv:2303.13375* (2023).

Luke Oakden-Rayner, Jared Dunnmon, Gustavo Carneiro, and Christopher Ré. 2020. Hidden stratification causes clinically meaningful failures in machine learning for medical imaging. In *Proceedings of the ACM Conference on Health, Inference, and Learning (CHIL)*. 151-159. https://doi.org/10.1145/3368555.3384468

Ziad Obermeyer, Brian Powers, Christine Vogeli, and Sendhil Mullainathan. 2019. Dissecting racial bias in an algorithm used to manage the health of populations. *Science* 366, 6464 (2019), 447-453. https://doi.org/10.1126/science.aax2342

Jean Ogier du Terrail, Samy-Safwan Ayed, Edwige Cyffers, Felix Grimberg, Chaoyang He, Regis Loeb, et al. 2022. FLamby: Datasets and Benchmarks for Cross-Silo Federated Learning in Realistic Healthcare Settings. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, Vol. 35.

Jean Ogier du Terrail, Armand Leopold, Clément Joly, Constance Béguier, Mathieu Andreux, Charles Maussion, et al. 2023. Federated learning for predicting histological response to neoadjuvant chemotherapy in triple-negative breast cancer. *Nature Medicine* 29 (2023), 135-146. https://doi.org/10.1038/s41591-022-02155-w

Cheng Ouyang, Carlo Biffi, Chen Chen, Turkay Kart, Huaqi Qiu, and Daniel Rueckert. 2020. Self-supervision with Superpixels: Training Few-Shot Medical Image Segmentation Without Annotation. In *European Conference on Computer Vision (ECCV) (LNCS, Vol. 12374)*. 762-780.

J. Marc Overhage, Patrick B. Ryan, Christian G. Reich, Abraham G. Hartzema, and Paul E. Stang. 2012. Validation of a common data model for active safety surveillance research. *Journal of the American Medical Informatics Association* 19, 1 (2012), 54-60. https://doi.org/10.1136/amiajnl-2011-000376

Kai Packhäuser, Sebastian Gündel, Nicolas Münster, Christopher Syben, Vincent Christlein, and Andreas Maier. 2022. Deep learning-based patient re-identification is able to exploit the biometric nature of medical chest X-ray data. *Scientific Reports* 12 (2022), 14851. https://doi.org/10.1038/s41598-022-19045-3

Suraj Pai, Dennis Bontempi, Ibrahim Hadzic, Vasco Prudente, Mateo Sokač, Tafadzwa L. Chaunzwa, et al. 2024. Foundation model for cancer imaging biomarkers. *Nature Machine Intelligence* 6 (2024), 354-367. https://doi.org/10.1038/s42256-024-00807-9

Sinno Jialin Pan and Qiang Yang. 2010. A Survey on Transfer Learning. *IEEE Transactions on Knowledge and Data Engineering* 22, 10 (2010), 1345-1359. https://doi.org/10.1109/TKDE.2009.191

Nicolas Papernot, Martín Abadi, Úlfar Erlingsson, Ian Goodfellow, and Kunal Talwar. 2017. Semi-supervised Knowledge Transfer for Deep Learning from Private Training Data. In *International Conference on Learning Representations (ICLR)*.

Sarthak Pati, Ujjwal Baid, Brandon Edwards, Micah Sheller, Shih-Han Wang, G. Anthony Reina, et al. 2022. Federated learning enables big data for rare cancer boundary detection. *Nature Communications* 13 (2022), 7346. https://doi.org/10.1038/s41467-022-33407-5

Peter Peduzzi, John Concato, Elizabeth Kemper, Theodore R. Holford, and Alvan R. Feinstein. 1996. A simulation study of the number of events per variable in logistic regression analysis. *Journal of Clinical Epidemiology* 49, 12 (1996), 1373-1379.

https://doi.org/10.1016/S0895-4356(96)00236-3

Maarten G. Poirot, Praneeth Vepakomma, Ken Chang, Jayashree Kalpathy-Cramer, Rajiv Gupta, and Ramesh Raskar. 2019. Split Learning for collaborative deep learning in healthcare. *arXiv preprint arXiv:1912.12115* (2019).

Viraj Prabhu, Anitha Kannan, Murali Ravuri, Manish Chaplain, David Sontag, and Xavier Amatriain. 2019. Few-Shot Learning for Dermatological Disease Diagnosis. In *Proceedings of the 4th Machine Learning for Healthcare Conference (MLHC) (PMLR, Vol. 106)*. 532-552.

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, et al. 2021. Learning Transferable Visual Models From Natural Language Supervision. In *Proceedings of the 38th International Conference on Machine Learning (ICML) (PMLR, Vol. 139)*. 8748-8763.

Maithra Raghu, Chiyuan Zhang, Jon Kleinberg, and Samy Bengio. 2019. Transfusion: Understanding Transfer Learning for Medical Imaging. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 32.

Jean Louis Raisaro, Juan Ramón Troncoso-Pastoriza, Mickaël Misbach, João Sá Sousa, Sylvain Pradervand, Edoardo Missiaglia, et al. 2019. MedCo: Enabling Secure and Privacy-Preserving Exploration of Distributed Clinical and Genomic Data. *IEEE/ACM Transactions on Computational Biology and Bioinformatics* 16, 4 (2019), 1328-1341. https://doi.org/10.1109/TCBB.2018.2854776

Maziar Raissi, Paris Perdikaris, and George E. Karniadakis. 2019. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *J. Comput. Phys.* 378 (2019), 686-707. https://doi.org/10.1016/j.jcp.2018.10.045

Laila Rasmy, Yang Xiang, Ziqian Xie, Cui Tao, and Degui Zhi. 2021. Med-BERT: pretrained contextualized embeddings on large-scale structured electronic health records for disease prediction. *npj Digital Medicine* 4 (2021), 86. https://doi.org/10.1038/s41746-021-00455-y

Alexander Ratner, Stephen H. Bach, Henry Ehrenberg, Jason Fries, Sen Wu, and Christopher Ré. 2017. Snorkel: Rapid Training Data Creation with Weak Supervision. *Proceedings of the VLDB Endowment* 11, 3 (2017), 269-282. https://doi.org/10.14778/3157794.3157797

Jenna M. Reps, Martijn J. Schuemie, Marc A. Suchard, Patrick B. Ryan, and Peter R. Rijnbeek. 2018. Design and implementation of a standardized framework to generate and evaluate patient-level prediction models using observational healthcare data. *Journal of the American Medical Informatics Association* 25, 8 (2018), 969-975. https://doi.org/10.1093/jamia/ocy032

Nicola Rieke, Jonny Hancox, Wenqi Li, Fausto Milletarì, Holger R. Roth, Shadi Albarqouni, et al. 2020. The future of digital health with federated learning. *npj Digital Medicine* 3 (2020), 119. https://doi.org/10.1038/s41746-020-00323-1

Richard D. Riley, Joie Ensor, Kym I. E. Snell, Frank E. Harrell, Glen P. Martin, Johannes B. Reitsma, Karel G. M. Moons, Gary Collins, and Maarten van Smeden. 2020. Calculating the sample size required for developing a clinical prediction model. *BMJ* 368 (2020), m441. https://doi.org/10.1136/bmj.m441

Richard D. Riley, Paul C. Lambert, and Ghada Abo-Zaid. 2010. Meta-analysis of individual participant data: rationale, conduct, and reporting. *BMJ* 340 (2010), c221. https://doi.org/10.1136/bmj.c221

Richard D. Riley, Kym I. E. Snell, Joie Ensor, Danielle L. Burke, Frank E. Harrell, Karel G. M. Moons, and Gary S. Collins. 2019a. Minimum sample size for developing a multivariable prediction model: Part I - Continuous outcomes. *Statistics in Medicine* 38, 7 (2019), 1262-1275. https://doi.org/10.1002/sim.7993

Richard D. Riley, Kym I. E. Snell, Joie Ensor, Danielle L. Burke, Frank E. Harrell, Karel G. M. Moons, and Gary S. Collins. 2019b. Minimum sample size for developing a multivariable prediction model: PART II - binary and time-to-event outcomes. *Statistics in Medicine* 38, 7 (2019), 1276-1296. https://doi.org/10.1002/sim.7992

Richard D. Riley, Kym I. E. Snell, Glen P. Martin, Rebecca Whittle, Lucinda Archer, Matthew Sperrin, and Gary S. Collins. 2021. Penalization and shrinkage methods produced unreliable clinical prediction models especially when sample size was small. *Journal of Clinical Epidemiology* 132 (2021), 88-96. https://doi.org/10.1016/j.jclinepi.2020.12.005

Michael Roberts, Derek Driggs, Matthew Thorpe, Julian Gilbey, Michael Yeung, Stephan Ursprung, et al. 2021. Common pitfalls and recommendations for using machine learning to detect and prognosticate for COVID-19 using chest radiographs and CT scans. *Nature Machine Intelligence* 3 (2021), 199-217. https://doi.org/10.1038/s42256-021-00307-0

Luc Rocher, Julien M. Hendrickx, and Yves-Alexandre de Montjoye. 2019. Estimating the success of re-identifications in incomplete datasets using generative models. *Nature Communications* 10 (2019), 3069. https://doi.org/10.1038/s41467-019-10933-3

Holger R. Roth, Ken Chang, Praveer Singh, Nir Neumark, Wenqi Li, Vikash Gupta, et al. 2020. Federated Learning for Breast Density Classification: A Real-World Implementation. In *Domain Adaptation and Representation Transfer, and Distributed and Collaborative Learning (DART/DCL) (LNCS, Vol. 12444)*. 181-191.

Maya Rotmensch, Yoni Halpern, Abdulhakim Tlimat, Steven Horng, and David Sontag. 2017. Learning a Health Knowledge Graph from Electronic Medical Records. *Scientific Reports* 7 (2017), 5994. https://doi.org/10.1038/s41598-017-05778-z

Abhijit Guha Roy, Shayan Siddiqui, Sebastian Pölsterl, Nassir Navab, and Christian Wachinger. 2019. BrainTorrent: A Peer-to-Peer Environment for Decentralized Federated Learning. *arXiv preprint arXiv:1905.06731* (2019).

Abhijit Guha Roy, Shayan Siddiqui, Sebastian Pölsterl, Nassir Navab, and Christian Wachinger. 2020. 'Squeeze & excite' guided few-shot segmentation of volumetric images. *Medical Image Analysis* 59 (2020), 101587. https://doi.org/10.1016/j.media.2019.101587

Francisco Sahli Costabal, Yibo Yang, Paris Perdikaris, Daniel E. Hurtado, and Ellen Kuhl. 2020. Physics-Informed Neural Networks for Cardiac Activation Mapping. *Frontiers in Physics* 8 (2020), 42. https://doi.org/10.3389/fphy.2020.00042

Oliver Lester Saldanha, Philip Quirke, Nicholas P. West, Jacqueline A. James, Maurice B. Loughrey, Heike I. Grabsch, et al. 2022. Swarm learning for decentralized artificial intelligence in cancer histopathology. *Nature Medicine* 28 (2022), 1232-1239. https://doi.org/10.1038/s41591-022-01768-5

Karthik V. Sarma, Stephanie Harmon, Thomas Sanford, Holger R. Roth, Ziyue Xu, Jesse Tetreault, et al. 2021. Federated learning improves site performance in multicenter deep learning without data sharing. *Journal of the American Medical Informatics Association* 28, 6 (2021), 1259-1264. https://doi.org/10.1093/jamia/ocaa341

Julia Schaefer, Moritz Lehne, Josef Schepers, Fabian Prasser, and Sylvia Thun. 2020. The use of machine learning in rare diseases: a scoping review. *Orphanet Journal of Rare Diseases* 15 (2020), 145. https://doi.org/10.1186/s13023-020-01424-6

Bernhard Schölkopf, Dominik Janzing, Jonas Peters, Eleni Sgouritsa, Kun Zhang, and Joris Mooij. 2012. On Causal and Anticausal Learning. In *Proceedings of the 29th International Conference on Machine Learning (ICML)*. 1255-1262.

Bernhard Schölkopf, Francesco Locatello, Stefan Bauer, Nan Rosemary Ke, Nal Kalchbrenner, Anirudh Goyal, and Yoshua Bengio. 2021. Toward Causal Representation Learning. *Proc. IEEE* 109, 5 (2021), 612-634. https://doi.org/10.1109/JPROC.2021.3058954

Christopher G. Schwarz, Walter K. Kremers, Terry M. Therneau, Richard R. Sharp, Jeffrey L. Gunter, Prashanthi Vemuri, et al. 2019. Identification of Anonymous MRI Research Participants with Face-Recognition Software. *New England Journal of Medicine* 381, 17 (2019), 1684-1686. https://doi.org/10.1056/NEJMc1908881

Burr Settles. 2009. *Active Learning Literature Survey*. Technical Report 1648. University of Wisconsin-Madison, Department of Computer Sciences.

Junyuan Shang, Cao Xiao, Tengfei Ma, Hongyan Li, and Jimeng Sun. 2019. GAMENet: Graph Augmented MEmory Networks for Recommending Medication Combination. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 33. 1126-1133.

Micah J. Sheller, Brandon Edwards, G. Anthony Reina, Jason Martin, Sarthak Pati, Aikaterini Kotrotsou, et al. 2020. Federated learning in medicine: facilitating multi-institutional collaborations without sharing patient data. *Scientific Reports* 10 (2020), 12598. https://doi.org/10.1038/s41598-020-69250-1

Micah J. Sheller, G. Anthony Reina, Brandon Edwards, Jason Martin, and Spyridon Bakas. 2019. Multi-institutional Deep Learning Modeling Without Sharing Patient Data: A Feasibility Study on Brain Tumor Segmentation. In *Brainlesion: Glioma, Multiple Sclerosis, Stroke and Traumatic Brain Injuries (BrainLes) (LNCS, Vol. 11383)*. 92-104.

Hoo-Chang Shin, Holger R. Roth, Mingchen Gao, Le Lu, Ziyue Xu, Isabella Nogues, Jianhua Yao, Daniel Mollura, and Ronald M. Summers. 2016. Deep Convolutional Neural Networks for Computer-Aided Detection: CNN Architectures, Dataset Characteristics and Transfer Learning. *IEEE Transactions on Medical Imaging* 35, 5 (2016), 1285-1298. https://doi.org/10.1109/TMI.2016.2528162

Hoo-Chang Shin, Neil A. Tenenholtz, Jameson K. Rogers, Christopher G. Schwarz, Matthew L. Senjem, Jeffrey L. Gunter, Katherine P. Andriole, and Mark Michalski. 2018. Medical Image Synthesis for Data Augmentation and Anonymization Using Generative Adversarial Networks. In *Simulation and Synthesis in Medical Imaging (SASHIMI) (LNCS, Vol. 11037)*. 1-11.

Reza Shokri, Marco Stronati, Congzheng Song, and Vitaly Shmatikov. 2017. Membership Inference Attacks Against Machine Learning Models. In *IEEE Symposium on Security and Privacy (S&P)*. 3-18. https://doi.org/10.1109/SP.2017.41

Connor Shorten and Taghi M. Khoshgoftaar. 2019. A survey on Image Data Augmentation for Deep Learning. *Journal of Big Data* 6 (2019), 60. https://doi.org/10.1186/s40537-019-0197-0

Ilia Shumailov, Zakhar Shumaylov, Yiren Zhao, Nicolas Papernot, Ross Anderson, and Yarin Gal. 2024. AI models collapse when trained on recursively generated data. *Nature* 631 (2024), 755-759. https://doi.org/10.1038/s41586-024-07566-y

Karan Singhal, Shekoofeh Azizi, Tao Tu, S. Sara Mahdavi, Jason Wei, Hyung Won Chung, et al. 2023. Large language models encode clinical knowledge. *Nature* 620 (2023), 172-180. https://doi.org/10.1038/s41586-023-06291-2

Asim Smailagic, Pedro Costa, Hae Young Noh, Devesh Walawalkar, Kartik Khandelwal, Adrian Galdran, et al. 2018. MedAL: Accurate and Robust Deep Active Learning for Medical Image Analysis. In *17th IEEE International Conference on Machine Learning and Applications (ICMLA)*. 481-488.

Virginia Smith, Chao-Kai Chiang, Maziar Sanjabi, and Ameet Talwalkar. 2017. Federated Multi-Task Learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 30.

Jake Snell, Kevin Swersky, and Richard Zemel. 2017. Prototypical Networks for Few-shot Learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 30.

Kihyuk Sohn, David Berthelot, Nicholas Carlini, Zizhao Zhang, Han Zhang, Colin A. Raffel, Ekin Dogus Cubuk, Alexey Kurakin, and Chun-Liang Li. 2020. FixMatch: Simplifying Semi-Supervised Learning with Consistency and Confidence.

In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 33.

Hari Sowrirajan, Jingbo Yang, Andrew Y. Ng, and Pranav Rajpurkar. 2021. MoCo Pretraining Improves Representation and Transferability of Chest X-ray Models. In *Proceedings of the Fourth Conference on Medical Imaging with Deep Learning (MIDL) (PMLR, Vol. 143)*.

Theresa Stadler, Bristena Oprisanu, and Carmela Troncoso. 2022. Synthetic Data - Anonymisation Groundhog Day. In *31st USENIX Security Symposium*. 1451-1468.

Charles Stein. 1956. Inadmissibility of the Usual Estimator for the Mean of a Multivariate Normal Distribution. In *Proceedings of the Third Berkeley Symposium on Mathematical Statistics and Probability*, Vol. 1. 197-206.

Ethan Steinberg, Ken Jung, Jason A. Fries, Conor K. Corbin, Stephen R. Pfohl, and Nigam H. Shah. 2021. Language models are an effective representation learning technique for electronic health record data. *Journal of Biomedical Informatics* 113 (2021), 103637. https://doi.org/10.1016/j.jbi.2020.103637

Ewout W. Steyerberg, Marinus J. C. Eijkemans, Frank E. Harrell, and J. Dik F. Habbema. 2000. Prognostic modelling with logistic regression analysis: a comparison of selection and estimation methods in small data sets. *Statistics in Medicine* 19, 8 (2000), 1059-1079.

Adarsh Subbaswamy, Peter Schulam, and Suchi Saria. 2019. Preventing Failures Due to Dataset Shift: Learning Predictive Models That Transport. In *Proceedings of the 22nd International Conference on Artificial Intelligence and Statistics (AISTATS) (PMLR, Vol. 89)*. 3118-3127.

Marc A. Suchard, Martijn J. Schuemie, Harlan M. Krumholz, Seng Chan You, RuiJun Chen, Nicole Pratt, et al. 2019. Comprehensive comparative effectiveness and safety of first-line antihypertensive drug classes: a systematic, multinational, large-scale analysis. *The Lancet* 394, 10211 (2019), 1816-1826. https://doi.org/10.1016/S0140-6736(19)32317-7

Latanya Sweeney. 2002. k-anonymity: A model for protecting privacy. *International Journal of Uncertainty, Fuzziness and Knowledge-Based Systems* 10, 5 (2002), 557-570. https://doi.org/10.1142/S0218488502001648

Canh T. Dinh, Nguyen Tran, and Josh Nguyen. 2020. Personalized Federated Learning with Moreau Envelopes. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 33.

Nima Tajbakhsh, Laura Jeyaseelan, Qian Li, Jeffrey N. Chiang, Zhihao Wu, and Xiaowei Ding. 2020. Embracing imperfect datasets: A review of deep learning solutions for medical image segmentation. *Medical Image Analysis* 63 (2020), 101693. https://doi.org/10.1016/j.media.2020.101693

Nima Tajbakhsh, Jae Y. Shin, Suryakanth R. Gurudu, R. Todd Hurst, Christopher B. Kendall, Michael B. Gotway, and Jianming Liang. 2016. Convolutional Neural Networks for Medical Image Analysis: Full Training or Fine Tuning? *IEEE Transactions on Medical Imaging* 35, 5 (2016), 1299-1312. https://doi.org/10.1109/TMI.2016.2535302

Aiham Taleb, Winfried Loetzsch, Danz Noel, Julius Severin, Thomas Gaertner, Benjamin Bergner, and Christoph Lippert. 2020. 3D Self-Supervised Methods for Medical Imaging. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 33.

Alysa Ziying Tan, Han Yu, Lizhen Cui, and Qiang Yang. 2023. Towards Personalized Federated Learning. *IEEE Transactions on Neural Networks and Learning Systems* 34, 12 (2023), 9587-9603. https://doi.org/10.1109/TNNLS.2022.3160699

Bernardo Camajori Tedeschini, Stefano Savazzi, Roman Stoklasa, et al. 2022. Decentralized Federated Learning for Healthcare Networks: A Case Study on Tumor Segmentation. *IEEE Access* 10 (2022), 8693-8708. https://doi.org/10.1109/ACCESS.2022.3141913

Christina V. Theodoris, Ling Xiao, Anant Chopra, Mark D. Chaffin, Zeina R. Al Sayed, Matthew C. Hill, et al. 2023. Transfer learning enables predictions in network biology. *Nature* 618 (2023), 616-624. https://doi.org/10.1038/s41586-023-06139-9

Robert Tibshirani. 1996. Regression Shrinkage and Selection via the Lasso. *Journal of the Royal Statistical Society: Series B* 58, 1 (1996), 267-288.

Ekin Tiu, Ellie Talius, Pujan Patel, Curtis P. Langlotz, Andrew Y. Ng, and Pranav Rajpurkar. 2022. Expert-level detection of pathologies from unannotated chest X-ray images via self-supervised learning. *Nature Biomedical Engineering* 6 (2022), 1399-1406. https://doi.org/10.1038/s41551-022-00936-9

Florian Tramèr and Dan Boneh. 2021. Differentially Private Learning Needs Better Features (or Much More Data). In *International Conference on Learning Representations (ICLR)*.

Nilesh Tripuraneni, Michael I. Jordan, and Chi Jin. 2020. On the Theory of Transfer Learning: The Importance of Task Diversity. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 33.

Akhil Vaid, Suraj K. Jaladanki, Jie Xu, Shelly Teng, Arvind Kumar, Samuel Lee, et al. 2021. Federated Learning of Electronic Health Records to Improve Mortality Prediction in Hospitalized Patients With COVID-19: Machine Learning Approach. *JMIR Medical Informatics* 9, 1 (2021), e24207. https://doi.org/10.2196/24207

Ben Van Calster, Maarten van Smeden, Bavo De Cock, and Ewout W. Steyerberg. 2020. Regression shrinkage methods for clinical prediction models do not guarantee improved performance: Simulation study. *Statistical Methods in Medical Research* 29, 11 (2020), 3166-3178. https://doi.org/10.1177/0962280220921415

Ruben van den Goorbergh, Maarten van Smeden, Dirk Timmerman, and Ben Van Calster. 2022. The harm of class imbalance corrections for risk prediction models: illustration and simulation using logistic regression. *Journal of the American*

Medical Informatics Association 29, 9 (2022), 1525-1534. https://doi.org/10.1093/jamia/ocac093

Maarten van Smeden, Karel G. M. Moons, Joris A. H. de Groot, Gary S. Collins, Douglas G. Altman, Marinus J. C. Eijkemans, and Johannes B. Reitsma. 2019. Sample size for binary logistic prediction models: Beyond events per variable criteria. *Statistical Methods in Medical Research* 28, 8 (2019), 2455-2474. https://doi.org/10.1177/0962280218784726

Paul Vanhaesebrouck, Aurélien Bellet, and Marc Tommasi. 2017. Decentralized Collaborative Learning of Personalized Models over Networks. In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS) (PMLR, Vol. 54)*. 509-517.

Gaël Varoquaux. 2018. Cross-validation failure: Small sample sizes lead to large error bars. *NeuroImage* 180 (2018), 68-77. https://doi.org/10.1016/j.neuroimage.2017.06.061

Gaël Varoquaux and Veronika Cheplygina. 2022. Machine learning for medical imaging: methodological failures and recommendations for the future. *npj Digital Medicine* 5 (2022), 48. https://doi.org/10.1038/s41746-022-00592-y

Praneeth Vepakomma, Otkrist Gupta, Tristan Swedish, and Ramesh Raskar. 2018. Split learning for health: Distributed deep learning without sharing raw patient data. *arXiv preprint arXiv:1812.00564* (2018).

Laura von Rueden, Sebastian Mayer, Katharina Beckh, Bogdan Georgiev, Sven Giesselbach, Raoul Heese, et al. 2023. Informed Machine Learning - A Taxonomy and Survey of Integrating Prior Knowledge into Learning Systems. *IEEE Transactions on Knowledge and Data Engineering* 35, 1 (2023), 614-633. https://doi.org/10.1109/TKDE.2021.3079836

Eugene Vorontsov, Alican Bozkurt, Adam Casson, George Shaikovski, Michal Zelechowski, Kristen Severson, et al. 2024. A foundation model for clinical-grade computational pathology and rare cancers detection. *Nature Medicine* 30, 10 (2024), 2924–2935. https://doi.org/10.1038/s41591-024-03141-0

Jason Walonoski, Mark Kramer, Joseph Nichols, Andre Quina, Chris Moesel, Dylan Hall, Carlton Duffett, Kudakwashe Dube, Thomas Gallagher, and Scott McLachlan. 2018. Synthea: An approach, method, and software mechanism for generating synthetic patients and the synthetic electronic health care record. *Journal of the American Medical Informatics Association* 25, 3 (2018), 230-238. https://doi.org/10.1093/jamia/ocx079

Xiaosong Wang, Yifan Peng, Le Lu, Zhiyong Lu, Mohammadhadi Bagheri, and Ronald M. Summers. 2017. ChestX-ray8: Hospital-scale Chest X-ray Database and Benchmarks on Weakly-Supervised Classification and Localization of Common Thorax Diseases. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 2097-2106.

Yanshan Wang, Sunghwan Sohn, Sijia Liu, Feichen Shen, Liwei Wang, Elizabeth J. Atkinson, Shreyasee Amin, and Hongfang Liu. 2019. A clinical text classification paradigm using weak supervision and deep representation. *BMC Medical Informatics and Decision Making* 19 (2019), 1. https://doi.org/10.1186/s12911-018-0723-6

Yaqing Wang, Quanming Yao, James T. Kwok, and Lionel M. Ni. 2020. Generalizing from a Few Examples: A Survey on Few-shot Learning. *Comput. Surveys* 53, 3 (2020), 63:1-63:34. https://doi.org/10.1145/3386252

Zifeng Wang, Zhenbang Wu, Dinesh Agarwal, and Jimeng Sun. 2022. MedCLIP: Contrastive Learning from Unpaired Medical Images and Text. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics, 3876–3887. https://doi.org/10.18653/v1/2022.emnlp-main.256

Stefanie Warnat-Herresthal, Hartmut Schultze, Krishnaprasad Lingadahalli Shastry, Sathyanarayanan Manamohan, Saikat Mukherjee, Vishesh Garg, et al. 2021. Swarm Learning for decentralized and confidential clinical machine learning. *Nature* 594 (2021), 265-270. https://doi.org/10.1038/s41586-021-03583-3

Kang Wei, Jun Li, Ming Ding, Chuan Ma, Howard H. Yang, Farhad Farokhi, Shi Jin, Tony Q. S. Quek, and H. Vincent Poor. 2020. Federated Learning With Differential Privacy: Algorithms and Performance Analysis. *IEEE Transactions on Information Forensics and Security* 15 (2020), 3454-3469. https://doi.org/10.1109/TIFS.2020.2988575

Martin J. Willemink, Wojciech A. Koszek, Cailin Hardell, Jie Wu, Dominik Fleischmann, Hugh Harvey, Les R. Folio, Ronald M. Summers, Daniel L. Rubin, and Matthew P. Lungren. 2020. Preparing Medical Imaging Data for Machine Learning. *Radiology* 295, 1 (2020), 4-15. https://doi.org/10.1148/radiol.2020192224

Andrew Wong, Erkin Otles, John P. Donnelly, et al. 2021. External Validation of a Widely Implemented Proprietary Sepsis Prediction Model in Hospitalized Patients. *JAMA Internal Medicine* 181, 8 (2021), 1065-1070. https://doi.org/10.1001/jamainternmed.2021.2626

Michael Wornow, Yizhe Xu, Rahul Thapa, Birju Patel, Ethan Steinberg, Scott Fleming, et al. 2023. The shaky foundations of large language models and foundation models for electronic health records. *npj Digital Medicine* 6 (2023), 135. https://doi.org/10.1038/s41746-023-00879-8

Yuan Wu, Xiaoqian Jiang, Jihoon Kim, and Lucila Ohno-Machado. 2012. Grid Binary LOgistic REgression (GLORE): building shared models without sharing data. *Journal of the American Medical Informatics Association* 19, 5 (2012), 758-764. https://doi.org/10.1136/amiajnl-2012-000862

Laure Wynants, Ben Van Calster, Gary S. Collins, Richard D. Riley, Georg Heinze, Ewoud Schuit, et al. 2020. Prediction models for diagnosis and prognosis of covid-19: systematic review and critical appraisal. *BMJ* 369 (2020), m1328. https://doi.org/10.1136/bmj.m1328

Lin Xiao and Stephen Boyd. 2004. Fast linear iterations for distributed averaging. *Systems & Control Letters* 53, 1 (2004), 65-78. https://doi.org/10.1016/j.sysconle.2004.02.022

An Xu, Wenqi Li, Pengfei Guo, Dong Yang, Holger R. Roth, Ali Hatamizadeh, Can Zhao, Daguang Xu, Heng Huang, and Ziyue Xu. 2022. Closing the Generalization Gap of Cross-silo Federated Medical Image Segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 20866-20875.

Hanwen Xu, Naoto Usuyama, Jaspreet Bagga, Sheng Zhang, Rajesh Rao, Tristan Naumann, et al. 2024. A whole-slide foundation model for digital pathology from real-world data. *Nature* 630 (2024), 181-188. https://doi.org/10.1038/s41586-024-07441-w

Jie Xu, Benjamin S. Glicksberg, Chang Su, Peter Walker, Jiang Bian, and Fei Wang. 2021. Federated Learning for Healthcare Informatics. *Journal of Healthcare Informatics Research* 5 (2021), 1-19. https://doi.org/10.1007/s41666-020-00082-4

Lei Xu, Maria Skoularidou, Alfredo Cuesta-Infante, and Kalyan Veeramachaneni. 2019. Modeling Tabular data using Conditional GAN. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 32.

Rui Yan, Liangqiong Qu, Qingyue Wei, Shih-Cheng Huang, Liyue Shen, Daniel L. Rubin, Lei Xing, and Yuyin Zhou. 2023. Label-Efficient Self-Supervised Federated Learning for Tackling Data Heterogeneity in Medical Imaging. *IEEE Transactions on Medical Imaging* 42, 7 (2023), 1932-1943. https://doi.org/10.1109/TMI.2022.3233574

Lin Yang, Yizhe Zhang, Jianxu Chen, Siyuan Zhang, and Danny Z. Chen. 2017. Suggestive Annotation: A Deep Active Learning Framework for Biomedical Image Segmentation. In *Medical Image Computing and Computer-Assisted Intervention (MICCAI) (LNCS, Vol. 10435)*. 399-407.

Qiang Yang, Yang Liu, Tianjian Chen, and Yongxin Tong. 2019. Federated Machine Learning: Concept and Applications. *ACM Transactions on Intelligent Systems and Technology* 10, 2 (2019), 12:1-12:19. https://doi.org/10.1145/3298981

Xin Yi, Ekta Walia, and Paul Babyn. 2019. Generative adversarial network in medical imaging: A review. *Medical Image Analysis* 58 (2019), 101552. https://doi.org/10.1016/j.media.2019.101552

Jinsung Yoon, Lydia N. Drumright, and Mihaela van der Schaar. 2020. Anonymization Through Data Synthesis Using Generative Adversarial Networks (ADS-GAN). *IEEE Journal of Biomedical and Health Informatics* 24, 8 (2020), 2378-2388. https://doi.org/10.1109/JBHI.2020.2980262

Alice C. Yu, Bahram Mohajer, and John Eng. 2022. External Validation of Deep Learning Algorithms for Radiologic Diagnosis: A Systematic Review. *Radiology: Artificial Intelligence* 4, 3 (2022), e210064. https://doi.org/10.1148/ryai.210064

Lequan Yu, Shujun Wang, Xiaomeng Li, Chi-Wing Fu, and Pheng-Ann Heng. 2019. Uncertainty-Aware Self-ensembling Model for Semi-supervised 3D Left Atrium Segmentation. In *Medical Image Computing and Computer-Assisted Intervention (MICCAI) (LNCS, Vol. 11765)*. 605-613.

John R. Zech, Marcus A. Badgeley, Manway Liu, Anthony B. Costa, Joseph J. Titano, and Eric Karl Oermann. 2018. Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs: A cross-sectional study. *PLoS Medicine* 15, 11 (2018), e1002683. https://doi.org/10.1371/journal.pmed.1002683

Hongyi Zhang, Moustapha Cisse, Yann N. Dauphin, and David Lopez-Paz. 2018. mixup: Beyond Empirical Risk Minimization. In *International Conference on Learning Representations (ICLR)*.

Haoran Zhang, Natalie Dullerud, Laleh Seyyed-Kalantari, Quaid Morris, Shalmali Joshi, and Marzyeh Ghassemi. 2021. An empirical framework for domain generalization in clinical settings. In *Proceedings of the ACM Conference on Health, Inference, and Learning (CHIL)*. 279-290. https://doi.org/10.1145/3450439.3451878

Xi Sheryl Zhang, Fengyi Tang, Hiroko H. Dodge, Jiayu Zhou, and Fei Wang. 2019. MetaPred: Meta-Learning for Clinical Risk Prediction with Limited Patient Electronic Health Records. In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*. 2487-2495. https://doi.org/10.1145/3292500.3330779

Yuhao Zhang, Hang Jiang, Yasuhide Miura, Christopher D. Manning, and Curtis P. Langlotz. 2022. Contrastive Learning of Medical Visual Representations from Paired Images and Text. In *Proceedings of the 7th Machine Learning for Healthcare Conference (MLHC) (PMLR, Vol. 182)*.

Ziqi Zhang, Chao Yan, Diego A. Mesa, Jimeng Sun, and Bradley A. Malin. 2020. Ensuring electronic medical record simulation through better training, modeling, and evaluation. *Journal of the American Medical Informatics Association* 27, 1 (2020), 99-108. https://doi.org/10.1093/jamia/ocz161

Amy Zhao, Guha Balakrishnan, Frédo Durand, John V. Guttag, and Adrian V. Dalca. 2019. Data augmentation using learned transformations for one-shot medical image segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 8543-8553.

Yue Zhao, Meng Li, Liangzhen Lai, Naveen Suda, Damon Civin, and Vikas Chandra. 2018. Federated Learning with Non-IID Data. *arXiv preprint arXiv:1806.00582* (2018).

Yukun Zhou, Mark A. Chia, Siegfried K. Wagner, Murat S. Ayhan, Dominic J. Williamson, Robbert R. Struyven, et al. 2023. A foundation model for generalizable disease detection from retinal images. *Nature* 622 (2023), 156-163. https://doi.org/10.1038/s41586-023-06555-x

Zongwei Zhou, Vatsal Sodha, Jiaxuan Pang, Michael B. Gotway, and Jianming Liang. 2021. Models Genesis. *Medical Image Analysis* 67 (2021), 101840. https://doi.org/10.1016/j.media.2020.101840

Ligeng Zhu, Zhijian Liu, and Song Han. 2019. Deep Leakage from Gradients. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 32.

Alexander Ziller, Dmitrii Usynin, Rickmer Braren, Marcus Makowski, Daniel Rueckert, and Georgios Kaissis. 2021. Medical imaging deep learning with differential privacy. *Scientific Reports* 11 (2021), 13524. https://doi.org/10.1038/s41598-021-93030-0